MSSI-Net: Multiscale Semantic-Guided Synergistic Interaction Network for Remote Sensing Image Change Detection

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***Abstract*—Remote sensing change detection (RSCD) has become an essential tool in observing and analyzing geographical information. However, existing deep learning approaches dependent solely on visual modalities may encounter challenges in discerning subtle variations amidst noise interference. To overcome these issues, we propose a multiscale semantic-guided synergistic interaction network (MSSI-Net), which utilizes the advanced multimodal semantic representations for enhancing the capacity to perceive hierarchical changes. Specifically, we first devise a multiscale interaction module (MIM) that leverages a multiscale attention mechanism to guide the interaction between the coarse and fine stages of different visual features. The fine-grained visual features subsequently complement the semantic features through scale weight reassignment to enhance the discriminative capability of vision-language features. Furthermore, driven by the semantic-guided synergistic interaction mechanism, our developed cross-modal feature fusion module (CFFM) exploits both homogeneous and heterogeneous features among modalities. This ensures that the generated vision-language features are semantically representative. Finally, we formulate a manifold differential perception head (MDPH) to optimize the detection of changes by efficiently fusing diverse differential feature representations, achieving comprehensive performance enhancement. Extensive experiments conducted on four benchmark datasets (LEVIR-CD, CDD, SYSU-CD, and WHU-CD) indicate that the designed MSSI-Net achieves state-of-the-art performance compared with existing methods.**

***Index Terms*—Change detection (CD), contrastive language-image pretraining (CLIP), remote sensing (RS).**

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I. Introduction

REMOTE sensing change detection (RSCD) is a crucial task in remote sensing (RS) technology, enabling the identification and quantification of temporal surface variations through comparative analysis of multitemporal imagery acquired over identical geographical regions [1], [2]. Driven by advancements in satellite observation systems and optical sensor technologies, RSCD has evolved into an essential analytical tool with applications spanning agricultural monitoring [3], environmental conservation [4], disaster response management [5], and urban development [6].

Conventional RSCD approaches depend on manually crafted features to identify land-cover alterations [7], [8]. Although these methods have achieved notable results in specific tasks, their effectiveness hinges critically on the quality of the indicators and the chosen thresholds, which fundamentally limits their applicability in very-high-resolution (VHR) RS images with intricate texture features and fine details [9]. In recent years, advancements in computer vision have prompted significant developments in CD methods, shifting toward more robust deep learning-based approaches. Convolutional neural networks (CNNs) such as fully convolutional network (FCN) [10] and U-Net [11] emerged as predominant solutions for extracting discriminative features from RS imagery, demonstrating strong capability in capturing local spatial dependencies and hierarchical contextual representations. Leveraging this powerful feature representation capacity, researchers have developed various CNN-based approaches that achieve state-of-the-art performance in RSCD tasks. For example, Peng et al. [12] developed a U-Net++ architecture incorporating multilateral branch fusion. Fang et al. [13] constructed the combination of Siamese network and Nested UNet (SNUNet) by integrating Siamese networks with NestedUNet. Daudt et al. [14] introduced three U-Net-based FCN variants: the feature-level based FC-Siam-Diff, the image-level-based FC-EF, and FC-Siam-Conc. The feature-level-based methods employ dual-weight-shared networks to extract temporally specific features separately, while image-level-based approaches process bitemporal images as composite inputs. Most subsequent studies have followed these two fundamental paradigms for bitemporal feature extraction [15], [16].

Subsequently, inspired by the remarkable performance of the transformer models [17], researchers have adopted specific

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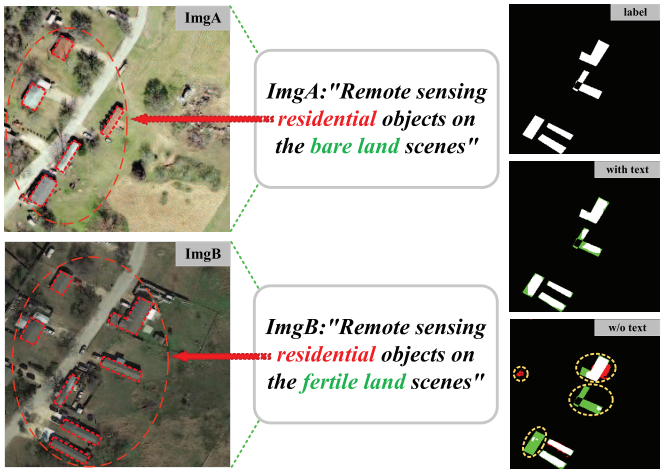


Fig. 1. Examples of how semantic guidance impacts the outcomes of RSCD. The small patches in red boxes represent the targets of interest, while the small patches in yellow circles represent noises or physical features not wanted in the result.

attention mechanisms, such as spatial-attention [18], [19], self-attention [20], [21], [22], and cross attention [23], [24], [25] to model long-range dependencies within deep features. Compared with conventional CNN architectures, transformer-based models demonstrate enhanced capacity for establishing global dependencies across entire images. In light of these works, Swin Transformer [26] significantly boosted its global modeling ability through the introduction of hierarchical representation and a self-attention mechanism grounded in moving windows, establishing a novel solution for CD. Based on the Swin-Transformer architecture, SwinSUNet [27] further captured spatial–temporal variations via multiscale feature fusion and detail preservation, attaining superior performance. Meanwhile, interaction and interscale feature fusion network (ICIF-Net) [28] integrated the strengths of CNNs and transformers, proposing mask-based aggregation and spatial alignment to fuse the capabilities of local feature extraction and global modeling, thereby endowing models with a global receptive field and opening new avenues for tasks requiring advanced semantic understanding.

Recently, there has been a growing interest in the development of foundation models [29], [30], [31], [32], [33], [34]. Empirical studies demonstrate that the fusion of vision and language significantly improves the interpretative capabilities of RS models while mitigating disruptions from single-modality biases [35], [36], [37], [38], [39]. As illustrated in Fig. 1, the model with semantic guidance improves the perception of changes in residential areas. This enhancement effectively reduces false positives and missed detections caused by seasonal and illumination variations. Consequently, it yields favorable detection results as shown in the "with text" image. Conversely, reliance solely on visual modality is more susceptible to such noise, resulting in relatively suboptimal outcomes as exemplified by the "w/o text" image. Nevertheless, most existing CNN and transformer-based networks primarily focus on visual recognition tasks, prioritizing the learning of image-level visual features while overlooking textual semantic understanding and lacking robust reasoning capabilities. Furthermore, these models typically demand substantial quantities of high-quality labeled image data to attain optimal performance, thereby limiting their generalization capacity for unseen data scenarios. Lately, the contrastive language-image pretraining (CLIP) foundation model [40] has exhibited powerful representation capability and robust zero-shot generalization talent through pretraining on a large-scale text–image paired dataset. This enables its application in diverse downstream tasks, including object retrieval [41], semantic segmentation [42], and object detection [43]. However, while the CLIP model excels at processing global context information, it still exhibits limitations when confronted with pixel-level fine-grained changes. Similarly, although textual descriptions can effectively convey given image information, high-resolution visual information in the RS domain often exceeds the textual descriptive capacity available to CLIP. The semantic disparity between various modalities may potentially obstruct the ability to precisely detect subtle alterations in VHR RS images, ultimately leading to decreased detection precision and accuracy.

To alleviate these challenges, we propose a CLIP-based multiscale semantic-guided synergistic interaction network (MSSI-Net) that effectively identifies differences in RS bitemporal images through enhanced multilevel semantic comprehension. Specifically, leveraging the visual-linguistic representation capacity of the RemoteCLIP model, we first generate semantic prompts for dual-temporal RS imagery. Subsequently, we develop the multiscale interaction module (MIM) to enhance the feature discriminability and enable further interaction between the coarse and fine stages of visual features. In addition, a cross-modal feature fusion module (CFFM) has been devised, which fuses semantic prompts with refined visual embeddings, thus assisting the model in comprehending both global contextual information and local details. Furthermore, we construct a manifold differential perception head (MDPH) that serves to optimize the recognition of bitemporal change features by fusing multiple differential feature representations. Extensive experiments on four benchmark datasets (LEVIR-CD, CDD, SYSU-CD, and WHU-CD) demonstrate that our MSSI-Net achieves state-of-the-art performance compared with existing methods.

In general, the key contributions of this article are outlined as follows.

1) With the collaborative support of RemoteCLIP, we propose a multi-scale semantic-guided interaction framework. This network can deeply mine multimodal semantic representations for boosting the capacity to perceive hierarchical changes while excluding noise interference, thus comprehensively improving the performance of RSCD.
2) An MIM is designed, which leverages a multilevel attention mechanism to guide the interaction between the coarse and fine stages of different visual features, thereby bolstering the discriminative capability of vision-language features.
3) We devise a CFFM that facilitates the hierarchical interactions between high-order semantic prompts and visual embeddings, aiding the model in comprehending both global contextual information and local details.

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4) An MDPH is suggested, which optimizes the perception of bitemporal change regions by seamlessly integrating diverse differential feature representations.

The remaining sections of this article is structured as follows. Section II gives a brief review of the most related work. In Section III, our proposed method is described in detail. Section IV shows the experimental results and analysis. In Section V, we presents a brief conclusion of our work.

II. Related Works

A. CNN-Based RSCD Methods

Deep learning-based approaches have shown substantial progress in the field of RSCD over the last several years [44]. Initially, CNN-based RSCD tasks were formulated as segmentation problems [14], where models similar to U-Net [11] achieved promising CD results by effectively integrating hierarchical features. Subsequently, the Siamese CNN architecture [45] was introduced to extract multitemporal features by processing paired images through parallel dual-branch networks [46]. For instance, Chen et al. [47] adopted a Siamese convolutional framework tailored for RSCD tasks, enabling the analysis of both structural and spatial–spectral information across multitemporal RS images to generate detailed change maps. Zhang and Shi [48] proposed a deep feature difference CNN incorporating a two-channel architecture with shared weights to generate multidepth and multiscale feature difference maps for CD, while integrating memory networks to encode temporal dependencies. On this basis, numerous subsequent studies emphasized multiscale feature learning through strategies including deep supervision [49], [50], and dense connections [51] to capture rich contextual information. Wu et al. [52] designed a joint-attention module for global guidance that incrementally aggregates multilevel differences via subtraction and concatenation operations, achieving notable performance. Fang et al. [53] implemented dense connections to integrate multiscale features and incorporated channel attention (CA) for thorough supervision, enabling precise CD. In addition, since CD aims to identify differences across multitemporal images, cross-scale feature interaction and fusion methods have recently been explored to obtain change-aware representations [54], [55], [56]. Although CNN-based approaches demonstrate superior performance in local feature extraction, they still struggle to integrate comprehensive global semantic information, thereby impeding its performance in CD tasks.

B. Attention-Based RSCD Methods

In order to alleviate the limitations of the receptive field in CNNs, the self-attention-based transformer architectures have been increasingly adopted for RSCD. For example, Chen et al. [57] configured a transformer-based network to accurately model the spatial–temporal contexts presented in bitemporal images, assisting the identification of changes while minimizing the influence of unrelated modifications. Bandara and Patel [58] employed a multilevel transformer encoder within a Siamese architecture. This setup was unified with a straightforward multilayer perceptron (MLP) decoder to deliver multiscale details, enabling precise CD. Although transformer models demonstrated potential in modeling long-range dependencies, they lack the necessary fine-grained details and local contextual guidance for precise localization, thereby impeding its performance in CD tasks. To address the above problem and accurately extract change features from VHR RS images, recent research works have explored the integration of parallel CNN and transformer architectures enhanced with attention-guided interaction mechanisms in CD tasks. Leveraging the complementary strengths of CNNs and transformers, Liu et al. [59] developed an attention-based multiscale transformer network that combines the strengths of CNNs, transformers, and attention mechanisms. Feng et al. [28] introduced the intrascale cross-ICIF-Net. This network employs a dual-branch architecture to extract multiscale detailed, and global features from bitemporal RS imageries. Ke and Zhang [60] presented a hybrid transformer module aimed at improving the quality of features extracted from a Siamese-based CNN backbone. The elevated features are subsequently harnessed by a cascade decoder for the prediction of the final change map. However, despite the previously mentioned methods having yielded significant advancements in CD, they are ineffective to process VHR images with intricate spatial patterns and diverse spectral characteristics due to insufficient contextual semantic comprehension [61], [62]. In addition, in practical applications, other high-resolution noise is unavoidably captured during the collection of RS data samples, which could result in additional computational burdens and interference.

C. Contrastive Language Image Pretraining

It is widely acknowledged that linguistic information inherently possesses richer semantic tapestry than visual data, offering the neural network multiple levels of comprehension [63]. To effectively utilize the abundant semantic information embedded within textual modality and seamlessly integrate it with visual data, it is imperative for us to introduce a method that is capable of delivering precise textual representations for VHR RS images. A prominent method that stands out in this regard is CLIP. The CLIP represents a revolutionary multimodal approach grounded in contrastive learning principles [64], [65], [66], its core idea is to construct positive and negative sample pairs from different views of the data and then aggregate positive and separate negative sample pairs in the feature space [67]. This principle facilitates the acquisition of robust features and ensures that similar data is closer in the feature space, enhancing the discriminative capability of the features. Unlike earlier visual recognition systems limited to recognizing the predefined categories, this method boosts remarkable generalization and extensibility, excelling in processing diverse image and text data without task-specific retraining [68].

Currently, it has demonstrated the versatility by achieving considerable success in a range of downstream tasks spanning multiple domains. For instance, Zhou et al. [69] employed the CLIP to obtain semantic information from different regions, which was subsequently used to inform the final predictive features. Xu et al. [70] trained a side network alongside CLIP to assist the model in local semantic perception, resulting in favorable outcomes. Xie et al. [71]

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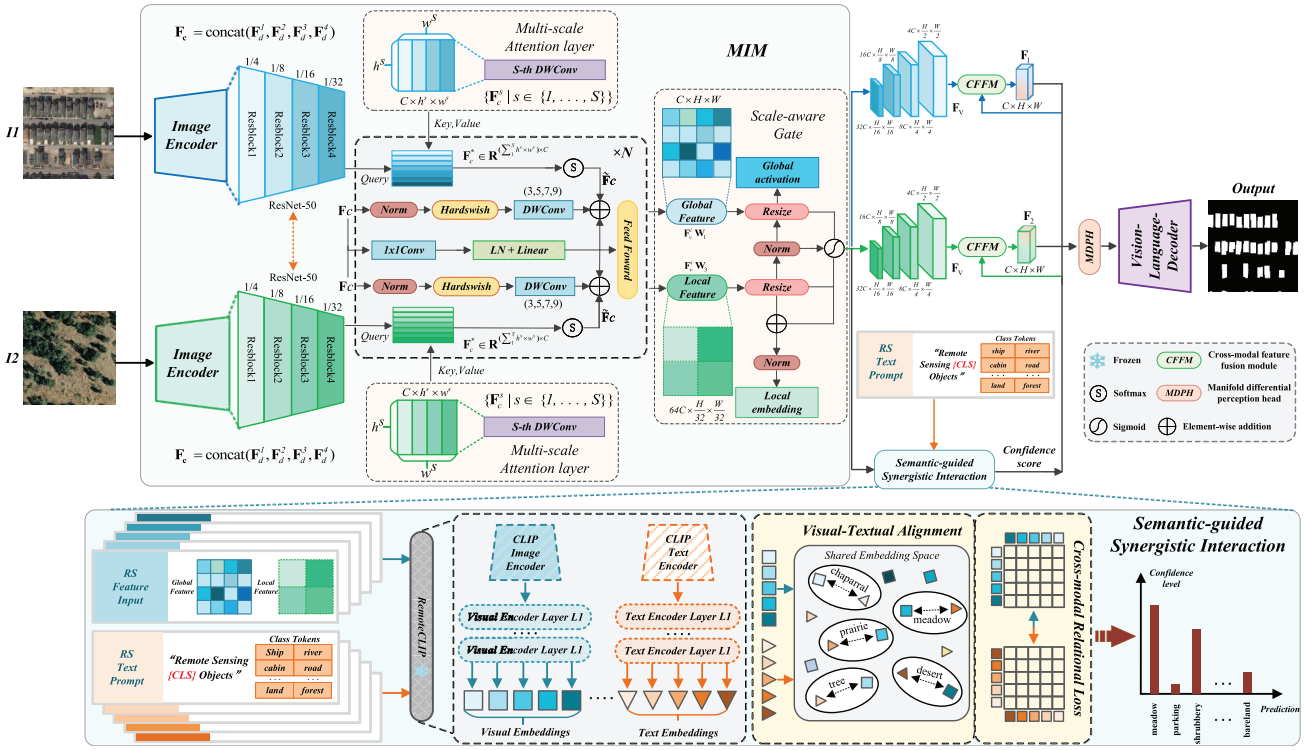


Fig. 2. Overall architecture of our MSSI-Net framework for CD. Our framework takes two inputs, prechange and postchange images, and predicts a binary semantic change map for the corresponding image pair. MSSI-Net consists of a vision-language driven encoder–decoder architecture, a pretraining foundation model RemoteCLIP, a feature enhancement MIM, a CFFM and spatial, and an MDPH for multivariate change feature integration.

introduced retrieval mechanisms and external knowledge bases into CLIP, obtaining superior results in several downstream tasks. Sun et al. [72] suggested that the precision of image recognition could be enhanced by integrating an additional alpha channel, which would aid the CLIP model to pinpoint the region of interest. Obinata et al. [73] devised a CLIP-based backend to refine CD by accurately perceiving semantic scene differences, thereby augmenting responses to environmental alterations. Liu et al. [74] established RemoteCLIP, the first vision-language foundation model for RS. Zhang et al. [75] further designed GeoRSCLIP, which has achieved significant performance gains on several RS-related tasks through the application of parameter-efficient fine-tuning techniques. In essence, employing CLIP within the realm of RSCD yields noteworthy advantages. By virtue of multimodal characteristics, it efficiently connects visual and textual data, thereby bolstering the ability to precisely interpret and adapt to intricate environmental alterations.

III. Proposed Method

A. Overall Architecture

Inspired by the preceding works, we propose an MSSI-Net RSCD network. This network leverages advanced multimodal semantic representations for discerning hierarchical alterations, ultimately achieving improved CD performance. As shown in Fig. 2, our framework is divided into five main parts: a backbone network consisting of a vision-language driven encoder–decoder architecture, a pretraining foundation model RemoteCLIP, a feature enhancement MIM, a CFFM, and an MDPH for multivariate change feature integration. Notably, we utilize the powerful visual-linguistic representational capability of the RemoteCLIP model to supervise the generation of higher order semantic prompts. Following that, the MIM performs further interactions between the coarse and fine stages of the visual features processed by the image encoder, rendering the features more discriminative. Subsequently, we develop a CFFM that merges high-order semantic prompts with refined visual embeddings to better help the model comprehend global contextual information and local details, which successfully overcome the limitations among traditional single-modal CD approaches. Moreover, our designed MDPH can effectively optimize the perception of bitemporal change features by efficiently fusing the disparity feature representations computed in various ways.

B. RemoteCLIP Model

RS images are distinguished by high spatial resolution, detailed multispectral information, and intricate background details. Whereas the pretraining dataset of RemoteCLIP comprises a substantial number and varied collection of RS images from numerous sources, including satellite imagery and autonomous aerial vehicle photography, among a broad spectrum of others. This diverse sample distribution enables the RemoteCLIP to accommodate more generalized feature representations, thereby improving its target category prediction accuracy and robustness across different scenarios. In this article, we introduce a method to derive higher order RS text prompts via RemoteCLIP. In particular, we leverage the sophisticated characterization ability of RemoteCLIP to predictively derive the corresponding textual representations

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for input dual-temporal RS images. This approach entails aligning highly correlated image–text samples and disambiguating mismatched ones. Formally, for a set of random image–text training sample pairs $\mathbf{x}$ and $\mathbf{t}$ that have been preprocessed, our primary objective is to identify potential correlation representations between different features obtained through encoding them with the image encoder F_{ve} and the text encoder F_{te}. Furthermore, we facilitate a network in cross-modal feature matching by exploiting the intrinsic prior RS knowledge of the RemoteCLIP. The process can be expressed as follows:

$$\mathbf{F}_v^i = F_{ve}\left(\mathbf{x}^i\right) \tag{1}$$

$$\mathbf{F}_t^i = F_{te}\left(\mathbf{t}^i\right) \tag{2}$$

$$\mathbf{Z}_{\text{aligned}} = \arg\max \mathbf{z}\left(\text{sim}\left(\mathbf{z}\right) \cdot \text{conf}\left(\mathbf{z}\right)\right) \tag{3}$$

$$\mathbf{Z}_{\text{separated}} = \mathbf{z} \mid \text{sim}\left(\mathbf{z}\right) \cdot \text{conf}\left(\mathbf{z}\right) < \tau \tag{4}$$

where $\mathbf{z}$ denotes $(\mathbf{F}_v^i, \mathbf{F}_t^i)$, sim$(\cdot)$ represents the similarity calculation, and conf$(\cdot)$ represents the confidence score. This process not only optimizes the quality of the semantic prompt obtained from the text encoder but also provides valuable high-confidence global semantic guidance for the subsequent cross-modal feature fusion process.

C. Multiscale Interaction Module

To tackle the multiscale variations in RS images and delve deeper into the synergistic benefits of visual and textual modalities, we have conceived an MIM. This module facilitates enhanced interaction among the multilevel visual features gathered at every stage of the image encoder. The refined visual embeddings obtained from MIM will engage in cross-modal interaction with the high-confidence semantic prompts derived from the RemoteCLIP model.

Concretely, the module initially performs a multilevel feature combination through downsampling of the disparate visual features $\mathbf{F}_v^i \in \mathbb{R}^{C_i \times H_i \times W_i}$, to a uniform size and subsequent concatenation along the channel dimension. This can be expressed as follows:

$$\mathbf{F}_d^i = \text{downsample}\left(\mathbf{F}_v^i\right), \quad i \in \{1, 2, 3, 4\} \tag{5}$$

$$\mathbf{F}_c = \text{concat}\left(\mathbf{F}_d^1, \mathbf{F}_d^2, \mathbf{F}_d^3, \mathbf{F}_d^4\right) \tag{6}$$

where $\mathbf{F}_d^i$ represents the downsampled features, $\mathbf{C}_1 = 4, \mathbf{C}_2 = 8, \mathbf{C}_3 = 16$, and $\mathbf{C}_4 = 32$. Whereas $\mathbf{F}_c$ represents the multistage feature concatenated along the channel dimension and the downsampling operation downsample$(\cdot)$ is typically implemented through average pooling. Furthermore, to realize the interaction of visual features at multiple scales, we conceive a multiscale attention layer. In addition, the concatenated features $\mathbf{F}_c \in \mathbb{R}^{C \times H \times W}$ will undergo deep convolution with varying step and kernel sizes. This process allows for adaptation in diverse perceptive fields, as detailed in the following equation:

$$\mathbf{F}_c^s = \mathbf{k}^s \cdot \mathbf{F}_c \tag{7}$$

$$h^s = \frac{h-1}{s} + 1, w^s = \frac{w-1}{s} + 1 \tag{8}$$

where $s \in \{1, \ldots, S\}$, S is the number of resized scales, $\mathbf{k}^s$ is the sth depthwise convolution, and h^s and w^s are the corresponding height and width of the $\mathbf{F}_c^s \in \mathbb{R}^{C \times h^s \times w^s}$, respectively. With the set $\{\mathbf{F}_c^s \mid s \in \{1, \ldots, S\}\}$, the elements on the size dimension are flattened and concatenated as a sequence feature, represented as $\mathbf{F}_c^* \in \mathbb{R}^{\left(\sum_1^S h^s \times w^s\right) \times C}$. Afterward, the origin feature $\mathbf{F}_c$ is taken as the query, whereas the multiscale-aware feature $\mathbf{F}_c^*$ is used as the key and value, enabling cross-scale interaction as follows:

$$\hat{\mathbf{F}}_c = \text{softmax}\left(\frac{\mathbf{F}_c \mathbf{W}_q \cdot \mathbf{F}_c^* \mathbf{W}_k^T}{\sqrt{C}}\right) \cdot \mathbf{F}_c^* \mathbf{W}_{\mathbf{v}}. \tag{9}$$

During this procedure, to ensure the optimal preservation of local details in the image, a local relation compensation, designated as LRC, is incorporated into the output component of the multiscale attention layer. In this way, the ultimate output of the multiscale attention layer is represented as follows:

$$\widetilde{\mathbf{F}}_{\mathbf{c}} = \hat{\mathbf{F}}_{\mathbf{c}} + \text{DWConv}\left(\text{Hardswish}\left(\mathbf{F}_c\right)\right) \tag{10}$$

where DWConv$(\cdot)$ represents the depthwise convolution and Hardswish$(\cdot)$ is the activity function, which is aimed to enhance the multiscale local information extraction. The feed-forward layer is structured in a manner that is analogous to the multiscale attention layer, which is identical to the standard attention block. The feature $\mathbf{F}_{\mathbf{c}}$ is partitioned into four components, each of which is upsampled to match the size of $\mathbf{F}_v^i$. These processed components are subsequently fed into the scale-aware gate, thereby yielding the definitive output $\widetilde{\mathbf{F}}_{\mathbf{c}}$.

The scale-aware gate is ultimately adopted to bridge the semantic gap that emerges before and after the application of multiscale attention. In particular, for each component of $\mathbf{F}_{\mathbf{c}}$, the corresponding element from F_{ve} is employed to quantify the influence of cross-scale interaction. The ultimate output of the scale-aware gate $\mathbf{F}_V \in \mathbb{R}^{C \times H \times W}$ is then utilized for cross-modal feature integration with higher order semantic prompts

$$\mathbf{F}_V = \sigma\left(\mathbf{F}_v^i \mathbf{W}_1\right) \odot \widetilde{\mathbf{F}}_{\mathbf{c}}^{\mathbf{i}} \mathbf{W}_2 + \mathbf{F}_v^i \mathbf{W}_3 \tag{11}$$

where $\sigma(\cdot)$ denotes the Sigmoid normalization and $\odot$ indicates the Hadamard product.

D. Higher Order Cross-Modal Feature Fusion Module

It is widely recognized that textual modalities can furnish a wealth of object semantic information, whereas the visual modal data provide rich visual details encompassing shape, color, texture, and dynamic changes. In light of the current limitations of RSCD methods in expressing change features across multitemporal scenarios and their inability to deliver fine-grained and robust change descriptions. We present a CFFM grounded in the higher order semantic guidance. As illustrated in Fig. 3, this module implements a dual-path architecture integrating Nth-order spatial–channel interactions for multiscale cross-modal fusion, fosters seamless information interaction between heterogeneous modalities. Where the spatial interaction branch employs the progressive refinement of cross-modal attention maps through iterative operations, each iteration applies frequency-domain-optimized self-attention to compute spatially aligned text–visual correlations via fast Fourier transform (FFT)-accelerated matrix

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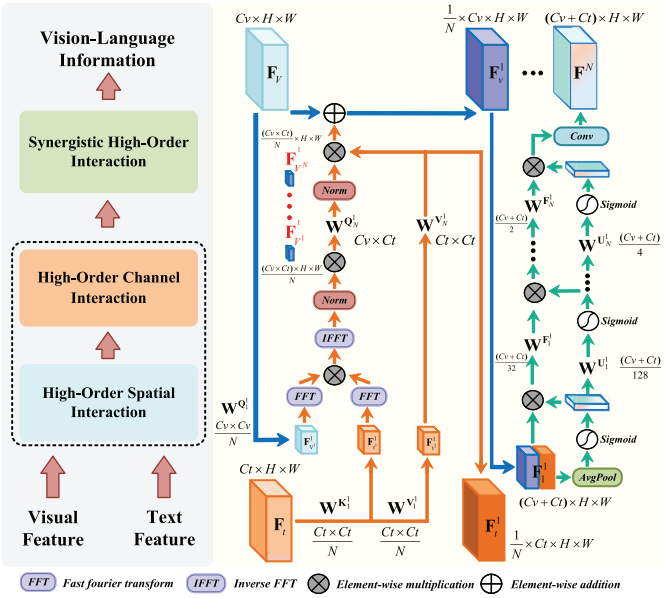


Fig. 3. Framework of the proposed CFFM. The spatial high-order interaction sufficiently elucidates collaborations between two modalities, integrating spatial fine-grained complementary information through the application of high-order modeling. The channel high-order interaction further distinguishing interdependencies between vision and test modalities.

multiplications. This process adaptively weights complementary features across modalities while suppressing noise through residual normalization. In addition, the channel interaction branch incorporates an enhanced squeeze-and-excitation mechanism, which performs dynamic channel recalibration through cascaded gating operations, amplifying discriminative cross-modal channel dependencies. The final fusion stage integrates spatially reweighted features and channel-recalibrated representations via depthwise separable convolutions, illustrating how initial coarse semantic alignments progressively evolve into fine-grained vision-language features through iterative optimization, thereby enabling comprehensive and precise interpretation of bitemporal RS changes.

To enhance multiscale spectral–spatial interactions between modalities, we embed FFT and inverse FFT (IFFT) into the iterative interaction framework. During spatial high-order interactions, FFT is applied to modality-aware features to decompose them into frequency spectra. A learnable frequency filter then modulates these spectra by suppressing noise-dominant high-frequency components, while enhancing task-relevant mid-frequency bands that capture texture details. This spectral filtering ensures cross-modal frequency synchronization, aligning dominant frequency patterns between text and vision features to enable phase-aware fusion of semantically correlated features. The refined frequency spectra are subsequently reconstructed via IFFT back to the spatial domain, yielding features that simultaneously preserve global contextual coherence and local discriminative details. This dual-domain transformation process that combined with iterative interactions, enables CFFM to adaptively integrate features across scales from holistic scene layouts to fine-grained object variations.

The high-order spatial interactions mechanism initiates a process of distinguishing and aggregating complementary information, utilizing the self-attention mechanism to focus on textual and visual features with precision and depth. Unlike conventional self-attentive modules limited to second-order spatial interactions, our approach is broadened to incorporate interactions of any order, all while preserving efficiency. Concretely, for each iteration, the following formulation is extended as:

$$\mathbf{F}_V^i = \text{Attn}\left(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i\right) \tag{12}$$

$$\mathbf{F}_{tj}^i = \mathbf{F}_{tj-1}^i \mathbf{W}^{\mathbf{K}_{j-1}^i} \tag{13}$$

$$\mathbf{F}_{Vj}^i = \text{Norm}\left(\mathbf{F}_{Vj-1}^i \mathbf{W}^{\mathbf{Q}_{j-1}^i}\right) \tag{14}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ denote the query, key and value, respectively, $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{C\times H\times W}$. Norm$(\cdot)$ represents the layer normalization, while $\mathbf{W}^{\mathbf{K}_{j-1}}$ and $\mathbf{W}^{\mathbf{Q}_{j-1}} \in \mathbb{R}^{C\times C}$ indicate the linear transformations applied to project modality-aware feature representations, $2 \le j \le N$. Both $\mathbf{F}_{tj}^i$ and $\mathbf{F}_{Vj}^i$ are projected into $\mathbb{R}^{C\times H\times W}$ through linear transformations, maintaining spatial and feature dimensions. The core of the high-order interaction for the jth order within the ith iteration is computed by modulating the projected features

$$\mathcal{S}_1\left(\left(\mathbf{F}_{Vj}^i\right)\right) = \left(\mathbf{F}_{Vj-1}^{i-1} \mathbf{W}^{\mathbf{Q}_{j-1}^i}\right) \odot \left(\mathbf{F}_{tj-1}^i \mathbf{W}^{\mathbf{V}_{j-1}^i}\right). \tag{15}$$

The value $\mathcal{S}_1(\cdot)$ signifies the output of the high-order spatial interaction, which captures spatial interactions of up to Nth-order related to the input features. The integration of diverse orders within each spatial interaction ensures the combination of complementary information across varying granularities, capturing a wide range of details from broad overviews to fine-grained specifics. Furthermore, the iterative nature of these interactions, where each iteration exhibits discriminatory responses tailored to the data nuances, progressively enriches the diversity of features. This iterative refinement not only amplifies the capacity to discern intricate patterns but also significantly enhances its ability to identify and leverage multiscale features.

The higher order channel interactions are modeled through the utilization of the advanced squeeze-and-excitation module, explicitly capturing the intricate interdependencies among input feature channels. In the context of fusing text and vision modalities, this module meticulously formulates the dependency between features derived from the ith spatial interaction, thereby facilitating a deeper and more nuanced integration of information from both modalities. The process can be formulated as follows:

$$\mathbf{U}^i = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \mathbf{F}^i(h, w) \tag{16}$$

where H and W denote the height and width of the output $\mathbf{F}^i = \text{concat}[\mathbf{F}_V^i, \mathbf{F}_t^i]$, respectively, and $\mathbf{U}^i \in \mathbb{R}^C$ represents the channelwise global average pooled feature vector. The channelwise dependencies are then iteratively refined as follows:

$$\mathbf{U}_{j-1}^i = \sigma\left(\mathbf{W}^{\mathbf{F}_{j-1}^i} \mathbf{U}_{j-2}^i\right) \tag{17}$$

$$\mathbf{F}_{j-1}^i = \mathbf{W}^{\mathbf{F}_{j-1}^i} \mathbf{F}_{j-2}^i \tag{18}$$

$$\mathcal{S}_2\left(\mathbf{F}_j^i\right) = \sigma\left(\mathbf{W}^{\mathbf{U}_j^i} \mathbf{U}_{j-1}^i\right) \cdot \left(\mathbf{W}^{\mathbf{F}_j^i} \mathbf{F}_{j-1}^i\right) \tag{19}$$

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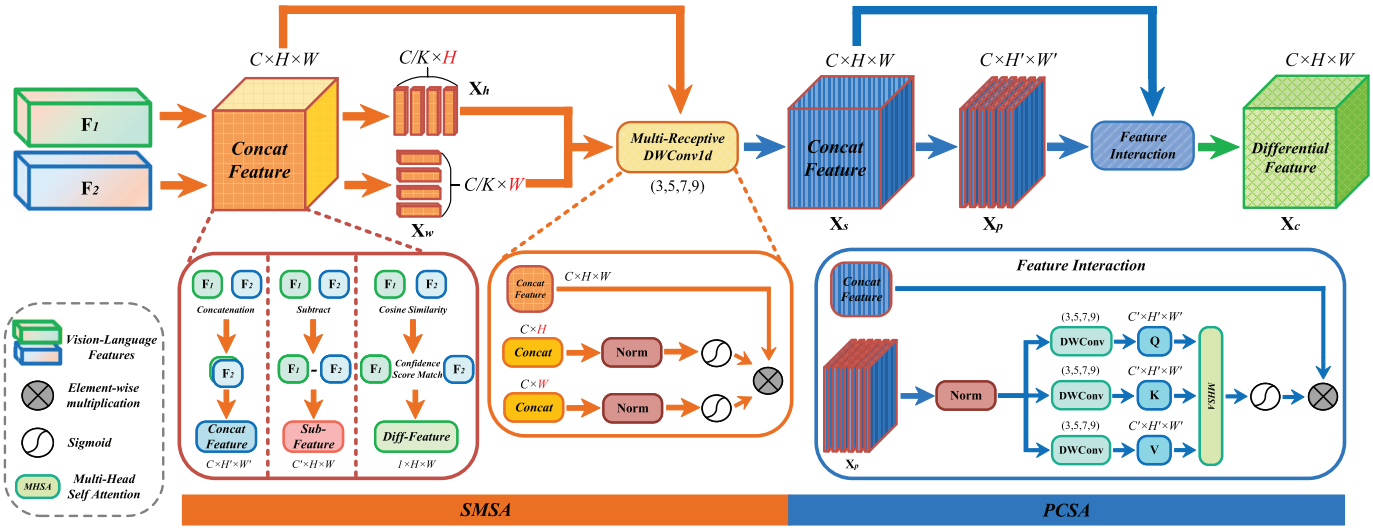


Fig. 4. Framework of the proposed MDPH. The Concat feature refers to the feature map obtained by the process of concatenation of the bitemporal images. The subtraction feature entails the subtraction of the feature maps of the bitemporal images. The term bitemporal difference map is used to describe a differential map, which is calculated by the cosine similarity.

where $\mathbf{W}^{\mathbf{F}_j^i} \in \mathbb{R}^{C\times C}$ is a learnable weight matrix, $\mathbf{W}^{\mathbf{U}_j^i} \in \mathbb{R}^{C\times C}$ comprises two linear transformations and a rectified linear unit (ReLU) activation function, and $\mathcal{S}_2(\cdot)$ signifies the output of the channel interaction. Ultimately, the fused features generated through convolutional processing will be employed to make final differential feature predictions at subsequent stage.

E. Manifold Differential Perception Head

In this section, we delve deeply into various variational feature extraction techniques and successfully unify them within a cohesive framework. The first approach is feature map subtraction, a technique that effectively facilitates the acquisition of change features. However, the resultant differential features may be impacted by the reduction of semantic details that were originally present in the individual maps. Furthermore, while the concatenation of feature maps preserves the comprehensive feature information, it fails to explicitly represent the differential features. In addition, we introduce a similarity-based computation method using the cosine function to describe the difference map of bitemporal features. Consequently, as shown in Fig. 4, for the given three inputs: subtraction difference feature map, bitemporal concat feature map, and bitemporal cosine similarity difference feature map, we propose an efficient MDPH to integrate semantic information from these three representations, allowing for adaptive extraction of the differential features. In particular, MDPH comprises two key components: a shared multiperspective spatial attention (SMSA) stage and a progressive channelwise self-attention (PCSA) stage. The SMSA initially leverages multiscale depth-shared 1-D convolution to capture the semantic spatial information of each feature map, thereby effectively integrating global context dependency and semantic spatial prior. The PCSA further serves to mitigate discrepancies in semantic information among various subfeatures within the framework of SMSA, leveraging robust feature interactions through a self-attention mechanism. This process enhances the key discriminative features while simultaneously facilitating the removal of redundant information.

First, we employ the feature pyramid network (FPN) model to integrate bitemporal vision-language features. Furthermore, we compute the absolute value of the subtraction result to better depict the saliency map of the region of interest obtained through the subtraction. Moreover, considering that cosine distance can represent the similarity of different feature vectors in high-dimensional space, we have also explored the method in depth and successfully integrate them for the subsequent adaptive difference feature extraction process. The procedure can be delineated as follows:

$$\mathbf{F}_{\text{sub}} = \text{conv}\left(\text{abs}\left(\mathbf{F}_1 - \mathbf{F}_2\right)\right) \tag{20}$$

$$\varphi = \frac{\mathbf{x}_1 \cdot \mathbf{x}_2}{\max\left(\|\mathbf{x}_1\|_2 \cdot \|\mathbf{x}_2\|_2\right)}, \mathbf{x}_1 \in \mathbf{F}_1, \mathbf{x}_2 \in \mathbf{F}_2 \tag{21}$$

$$\mathbf{m} = \sigma\left(1 - \varphi\left(\mathbf{F}_1, \mathbf{F}_2\right)\right) \tag{22}$$

$$\mathbf{X} = \text{FPN}\left(\text{concat}\left(\mathbf{F}_1, \mathbf{F}_2\right)\right) \tag{23}$$

$$\mathbf{X} = \text{CA}\left(\text{concat}\left(\mathbf{X} \times \mathbf{m}, \mathbf{F}_{\text{sub}}, \mathbf{X}\right)\right) \tag{24}$$

where $\mathbf{F}_1$, $\mathbf{F}_2 \in \mathbb{R}^{C\times H\times W}$ represent the feature maps that encapsulate the vision-language information of temporal periods, $\mathbf{F}_{\text{sub}} \in \mathbb{R}^{C'\times H\times W}$ denotes the subtraction difference of the feature map, φ signifies the cosine similarity calculation, $\mathbf{m} \in \mathbb{R}^{1\times H\times W}$ is the difference map calculated by the cosine similarity map of the bitemporal feature maps, and $\mathbf{X}$ is defined as the Concat Feature that integrates the above three types of differential features. In addition, CA is the channel attention module to merge these three differential feature representations, enabling adaptive control over the significance of differential features. This attention mechanism allows the network to automatically adjust the weight distribution of the differential features according to the importance of altered areas, thereby enhancing the perceptual capabilities for changed features.

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Subsequently, the SMSA stage enables effective guidance by integrating multilevel spatial semantic information. Specifically, we decompose the given input feature map $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ along the height and width dimensions, and subsequently apply global average pool to each dimension, thereby creating two unidirectional 1-D sequence structures $\mathbf{X}_H \in \mathbb{R}^{C\times W}$ and $\mathbf{X}_W \in \mathbb{R}^{C\times H}$. Furthermore, to learn different spatial distributions and contextual relationships, we partition the feature set into K identically sized independent subfeatures $\mathbf{X}_H^i \in \mathbb{R}^{C/K\times W}$ and $\mathbf{X}_W^i \in \mathbb{R}^{C/K\times H}$, with a channel count of C/K for each subfeature. The process is presented as follows:

$$\mathbf{X}_H^i = \mathbf{X}_H\left[:, (i-1)\cdot\frac{C}{K} : i\cdot\frac{C}{K}, :\right] \tag{25}$$

$$\mathbf{X}_W^i = \mathbf{X}_W\left[:, (i-1)\cdot\frac{C}{K} : i\cdot\frac{C}{K}, :\right] \tag{26}$$

where $\mathbf{X}^i$ represents the ith subfeature, $i \in \{1, \ldots, K\}$. It is worth mentioning that each subfeature is independent, facilitating efficient extraction of multiperspective spatial information. Following the division of the feature set into independent subfeatures, we apply a range of scale kernel sizes in the subfeatures for depth-sharing 1-D convolution for alignment, enhancing consistency between features, and minimizing the semantic gap. This procedure can be formulated using the following equation:

$$\tilde{\mathbf{X}}_H^i = \text{DWConv } 1d_{k_i}^{\frac{C}{K}}\left(\mathbf{X}_H^i\right) \tag{27}$$

$$\tilde{\mathbf{X}}_W^i = \text{DWConv } 1d_{k_i}^{\frac{C}{K}}\left(\mathbf{X}_W^i\right) \tag{28}$$

where $\tilde{\mathbf{X}}^i$ means the spatial structural information of the ith subfeature obtained after convolutional operations. k_i denotes the convolutional kernel implemented on the ith subfeature.

We then perform subfeature aggregation through the application of group normalization. This model allows for the independent normalization of each subfeature, ensuring that no statistical noise is introduced. By doing so, it adeptly mitigates semantic interference among subfeatures and effectively prevents attentional dilution. Finally, the spatial attention can be obtained by leveraging a simple Sigmoid activation function to activate and suppress specific spatial regions, providing an accurate spatial prior for the computation of CA as follows:

$$\begin{aligned}\text{Attn}_H &= \sigma\left(\text{GN}_H^K\left(\text{Concat}\left(\tilde{\mathbf{X}}_H^1, \ldots, \tilde{\mathbf{X}}_H^K\right)\right)\right)\\ \text{Attn}_W &= \sigma\left(\text{GN}_W^K\left(\text{Concat}\left(\tilde{\mathbf{X}}_W^1, \ldots, \tilde{\mathbf{X}}_W^K\right)\right)\right)\end{aligned} \tag{29}$$

$$\mathbf{X}_s^i = \text{Attn}_H \cdot \text{Attn}_W \cdot \mathbf{X} \tag{30}$$

where $\text{Attn}_H \in \mathbb{R}^{C\times W}$ and $\text{Attn}_W \in \mathbb{R}^{C\times H}$ denote the heightwise and widthwise attention maps, respectively. $\text{GN}_H^K(\cdot)$ and $\text{GN}_W^K(\cdot)$ represent the group normalization with K groups along the H and W dimensions, respectively.

The PCSA stage further mitigates the semantic differences between various subfeatures in SMSA by utilizing self-attention-based feature interaction. Concretely, to efficiently calculate CA, the spatial information of the feature map is primarily aggregated through the operation of average pooling and group normalization. Following this, the overall feature mapping $\mathbf{F}_{\text{proj}}$ is achieved via applying depthwise separable 1-D convolution. The process can be expressed as follows:

$$\mathbf{X}_p^i = \text{Pool}^{(H,W)\to(H',W')}\left(\mathbf{X}_s^i\right) \tag{31}$$

$$\mathbf{F}_{\text{proj}} = \text{DWConv } 1\ d_{(1,1)}^{C\to C} \tag{32}$$

where $\text{Pool}(\cdot)$ denotes the pooling operation that rescales the resolution from (H, W) to (H', W'). $\mathbf{F}_{\text{proj}} \in \mathbb{R}^{C'\times H'\times W'}$ represents the mapping function. Subsequently, we employ a multihead self-attention mechanism in conjunction with spatial prior semantic information derived from SMSA to refine the semantic comprehension of local subfeatures. This is achieved by computing interchannel similarity through the overall feature mapping, which reduces the semantic discrepancies induced by multiscale convolution within the SMSA module. The procedure is defined as follows:

$$\mathbf{Q} = \mathbf{F}_{\text{proj}}^{\mathbf{Q}}\left(\mathbf{X}_p\right), \mathbf{K} = \mathbf{F}_{\text{proj}}^{\mathbf{K}}\left(\mathbf{X}_p\right), \mathbf{V} = \mathbf{F}_{\text{proj}}^{\mathbf{V}}\left(\mathbf{X}_p\right) \tag{33}$$

$$\mathbf{X}_{\text{attn}} = \text{Attn}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{C}}\right)\mathbf{V} \tag{34}$$

$$\mathbf{X}_c = \mathbf{X}_s^i \cdot \sigma\left(\text{Pool}_{(H',W')}^{(H',W')\to(1,1)}\left(\mathbf{X}_{\text{attn}}\right)\right). \tag{35}$$

Finally, semantic alterations in the specified region are augmented by the mean pooling and the Sigmoid activation functions, while the presence of noise is mitigated. Compared with traditional techniques, MDPH displays a more robust capacity to comprehend input perception and is capable of exploiting multilevel spatial semantic knowledge to guide differential feature learning. The aggregated differential features obtained by this module will be input into the vision-language decoder for final change prediction.

F. Loss Function

For the optimization of our proposed MSSI-Net, the cross-entropy loss function is adopted. Precisely, the network is trained to categorize each pixel within the input imagery. The training process minimizes the discrepancy between the forecast map and the ground truth, distinguishing the pixels as either changed or unchanged. The underlying mathematical formula is as follows:

$$L_{\text{CE}} = -\frac{1}{H\times W}\sum_{h=1,w=1}^{H,W}\mathbf{Y}\left(h,w\right)\log\left(\hat{\mathbf{Y}}\left(h,w\right)\right) \tag{36}$$

where $\mathbf{Y}$ denotes the ground truth. $\mathbf{Y}(h, w)$ and $\hat{\mathbf{Y}}(h, w)$ are used to represent the label and probability prediction for the pixel situated at the coordinates (h, w), respectively.

IV. Experiments

A. Datasets

1) LEVIR-CD dataset [76] is centered on detecting changes in building areas and contains a total of 637 pairs of bitemporal RS images sourced from Google Earth. These images are accompanied by annotations for over 31 333 instances and encompass a variety of building types, including villas, high-rise apartments, small garages, and large warehouses. Each image in a pair measures 1024 × 1024 pixels and has a spatial resolution of 0.5 m/pixel. For consistent evaluation, these

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Algorithm 1 Forward Propagation of MSSI-Net

Input : A pair of bitemporal images $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2\}$
Output: The predicted change mask $\hat{Y}$

Initialize $RemoteCLIP$ with pre-trained weights;
$\mathbf{T} = \{\mathbf{t}_1, \mathbf{t}_2\} \leftarrow RemoteCLIP_{\text{image-text}}(\mathbf{x}_i)$;
for $i \in \{1, 2\}$ **do**
  $\mathbf{z} = (\mathbf{F}_v^i, \mathbf{F}_t^i) \leftarrow \text{Encoder}(\mathbf{x}_i, \mathbf{t}_i)$;
  $\mathbf{Z}_{\text{aligned}} \leftarrow \arg\max \mathbf{z}(\text{sim}(\mathbf{z}) \cdot \text{conf}(\mathbf{z}))$;
  $\mathbf{Z}_{\text{separated}} \leftarrow \{\mathbf{z} \mid \text{sim}(\mathbf{z}) \cdot \text{conf}(\mathbf{z}) < \tau\}$;
  // Multiscale Interaction Module
  for $j \in \{1, 2, 3, 4\}$ **do**
    $\mathbf{F}_d^j \leftarrow \text{AvgPool}(\mathbf{F}_v^j)$;
    $\mathbf{F}_c \leftarrow \text{Concat}(\mathbf{F}_d^1, \ldots, \mathbf{F}_d^4)$;
  $\hat{F}_c \leftarrow \text{softmax}(\frac{\mathbf{F}_c \mathbf{W}_q \cdot \mathbf{F}_c^* \mathbf{W}_k^T}{\sqrt{C}}) \cdot \mathbf{F}_c^* \mathbf{W}_v$;
  $\widetilde{\mathbf{F}}_c \leftarrow \hat{F}_c + \text{DWConv}(\text{Hardswish}(\mathbf{F}_c))$;
  $\mathbf{F}_V^i \leftarrow \sigma(\mathbf{F}_v^i \mathbf{W}_1) \odot (\widetilde{\mathbf{F}}_c \mathbf{W}_2) + \mathbf{F}_v^i \mathbf{W}_3$;
// Cross-modal Feature Fusion
for $n = 1$ **to** N **do**
  $\mathcal{S}_1 \leftarrow (\mathbf{F}_{V^{n-1}}^{j-1} \mathbf{W}^{\mathbf{Q}_{\mathbf{n-1}}^j}) \odot (\mathbf{F}_{t^{n-1}}^j \mathbf{W}^{\mathbf{V}_{\mathbf{n-1}}^j})$;
  $\mathcal{S}_2 \leftarrow \sigma(\mathbf{W}^{\mathbf{U}_n^j} \mathbf{U}_{n-1}^j) \cdot (\mathbf{W}^{\mathbf{F}_n^j} \mathbf{F}_{n-1}^j)$;
// Manifold Differential Perception
$\mathbf{X} \leftarrow [\Delta_{\text{sub}}, \Delta_{\text{concat}}, \Delta_{\cos}]$;
for $m = 1$ **to** M **do**
  $\text{Attn}_H \leftarrow \sigma(GN_H^m(\text{concat}(\tilde{X}_H^1, \ldots, \tilde{X}_H^M)))$;
  $\text{Attn}_W \leftarrow \sigma(GN_W^m(\text{concat}(\tilde{X}_W^1, \ldots, \tilde{X}_W^M)))$;
  $\mathbf{X}_s^m \leftarrow \text{Attn}_H \cdot \text{Attn}_W \cdot X$;
  $\mathbf{X}_p^m \leftarrow \text{Pool}^{(H,W)\to(H',W')}(\mathbf{X}_s^m)$;
  $\mathbf{X}_c \leftarrow \mathbf{X}_s^m \cdot \sigma(\text{Pool}_{(H',W')}^{(H',W')\to(1,1)}(\mathbf{X}_{\text{attn}}))$;
$\hat{Y} \leftarrow \text{Decoder}(\mathbf{X}_c)$

image pairs were cropped to a size of 256 × 256 pixels and have been divided into three subsets: a training set with 7120 pairs, a validation set with 1024 pairs, and a testing set with 2048 pairs, respectively.

2) CDD dataset [77] encompasses a wide range of changes, including urban development, vegetation alterations, and variations in water bodies. It comprises 11 pairs of dual-temporal, multisource RS images from Google Earth, with resolutions spanning from 0.03 to 1 m/pixel. For our experiments, these image pairs were cropped to a size of 256 × 256 pixels, and the dataset was subsequently divided into three subsets: a training set with 10 000 images, a validation set with 3000 images, and a testing set with 3000 images, respectively.
3) SYSU-CD dataset [78] comprises 20 000 pairs of high-resolution aerial images, each measuring 256 × 256 pixels and captured at the resolution of 0.5 m. These images were taken in Hong Kong between 2007 and 2014 and showcase a variety of changes in complex scenarios, including buildings, vegetation, croplands, roads, and sea. For our experimental purposes, the dataset is divided into three subsets: a training set with 12 000 image pairs, a validation set with 4000 image pairs, and a testing set with 4000 image pairs, respectively.
4) WHU-CD dataset [79] focuses on urban building CD and includes a pair of high-resolution RS images captured in the same area in 2012 and 2016, covering diverse building types such as residential areas, commercial structures, and industrial facilities. Each image measures 32 507 × 15 354 pixels, with a spatial resolution of 0.75 m. For standardized evaluation, the original images were cropped into 256 × 256 pixel patches and divided into three subsets: a training set containing 5947 patches, a validation set with 743 patches, and a test set comprising 744 patches, respectively.

B. Evaluation Metrics

To quantitatively evaluate the proposed method, we employ six evaluation metrics: precision (Pre) [80], recall (Rec) [81], intersection over union (IoU) [82], overall accuracy (OA) [83], $F1$-score [84] and Kappa coefficient (Kappa) [85]. Precision signifies the proportion of correctly results to all predicted changed samples, whereas recall quantifies the ratio of actual changed samples that are predicted as changed. In a task with many variables, higher precision and recall values correspond to reduced false positives and false negatives, respectively. Furthermore, their harmonic mean yields the $F1$-score, balancing these complementary metrics. In addition, in the CD task, IoU functions as a standard metric to assess the level of overlap between the predicted altered region and the actual ground truth, while OA represents the proportion of correctly classified samples out of the total samples. Moreover, the Kappa coefficient is utilized to quantify the agreement between model predictions and GT, incorporating adjustments for random classification to ensure a more robust evaluation of model performance.

C. Implementation Details

The implementation of MSSI-Net, along with all related experiments presented in this article, are developed using the PyTorch framework. All training and testing processes are carried out on NVIDIA GeForce RTX 4090 and NVIDIA RTX A100 GPUs. For model optimization, the AdamW optimizer is employed, configured with a learning rate of 0.0001 and a weight decay parameter set to 0.01. During the training phase, the cross-entropy loss function is utilized, computed solely on the decoder output. The batch size is set to 8, and the model is trained for a total of 20 000 iterations. Validation is conducted every 1000 iterations, and the model that demonstrates the best performance on the validation set is chosen for evaluation on the test set. Unless otherwise specified, the Resnet-50 from the CLIP model is used as the visual encoding network for the model, and attention-based residual integration system (ARIS) is enabled by default.

D. Comparison With State-of-the-Art Methods

To demonstrate the performance of our proposed method, we make a comprehensive comparison on four datasets (LEVIR-CD, CDD, SYSU-CD, and WHU-CD) between the MSSI-Net and the following 11 state-of-the-art methods, including five CNN and attention-based methods: improved

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separability network (ISNet) [86], dual-branch multilevel intertemporal network (DMINet) [52], lightweight Siamese network (LSNet) [87], Siamese network and NestedUNet (SNUNet) [13], and bitemporal feature alignment (BiFA) model [88]. Three transformer-based methods: Changeformer [58], SwinSUNet [27], and cross-temporal interaction with difference refinement network (CTIDRNet) [90]. Three CNN and transformer hybrid methods: bitemporal image transformer (BIT) [57], intrascale cross ICIF-Net [28], and domain distribution reconstruction learning (DDRL) [89]. A concise overview of these methods is provided in the following.

1) *ISNet [86]:* It is a CNN-based approach that leverages edge modules and bidirectional attention aggregation to effectively tackle challenges in change detection.
2) *DMINet [52]:* The network introduces a joint-attention block for global guidance and a robust architecture that incrementally aggregates multilevel differences via subtraction and concatenation.
3) *LSNet [87]:* A pioneering LSNet for RS change detection, efficiently extracting bitemporal features using shared weights.
4) *SNUNet [13]:* This is a network architecture tailored for CD, which employs dense connections to integrate multilevel features and incorporates CA for thorough supervision.
5) *ChangeFormer [58]:* A transformer-based multiscale feature connection method, which also combines an MLP decoder.
6) *SwinSUNet [27]:* It is a network based on the Swin Transformer, which captures spatial–temporal variations through multiscale feature fusion and detail preservation.
7) *BIT [57]:* The network efficiently integrates CNN and transformer. It converts the input image into semantic tags encapsulating high-level concepts of regions of interest and employs transformers to represent the contextual information.
8) *ICIF-Net [28]:* It utilizes a dual-backbone network consisting of CNN and transformer for intrascale interaction and interscale feature fusion, which strengthens interaction and fusion of different feature mappings.
9) *BiFA [88]:* It proposes a novel feature alignment method that achieves dual-temporal feature alignment at both temporal and multiscale levels to identify regions of interest while excluding extraneous changes.
10) *DDRL [89]:* This method leverages feature reconstruction and contrastive strategies to optimize low-level feature distributions and discriminability, thereby enhancing the change detection accuracy.
11) *CTIDRNet [90]:* The network employs hybrid attention for multilevel feature exploration and integrates an adaptive feature fusion module to improve the detection accuracy across diverse change categories.

E. Comparison and Analysis

1) Experimental Comparisons and Analysis on the LEVIR-CD Dataset: Table I displays the quantitative performance metrics attained by MSSI-Net and the compared methods on the LEVIR-CD dataset. The results indicate that our network outperforms other methods in terms of OA, IoU, $F1$, precision, and recall with 99.39%, 85.21%, 92.15%, 93.68%, and 90.67%, respectively. In comparison to recent methodologies, ISNet exhibits degraded performance in complex backgrounds, which is attributable to inadequate temporal interaction in its attention mechanism. DMINet attains IoU of 82.99% and $F1$ of 90.71% by leveraging intertemporal joint-attention blocks, underscoring the critical role of temporal feature alignment. In addition, ChangeFormer displays IoU and $F1$ of 82.66% and 90.50%, respectively, showcasing competitive results but are constrained by limited local detail modeling, leading to false detections in dense areas. DDRL distinguishes itself with a domain distribution reconstruction framework, achieving the second-highest scores through graph attention and contrastive learning. The superiority of our MSSI-Net stems from the MIM, which orchestrates hierarchical feature aggregation via multilevel attention mechanisms, particularly effective in scenarios containing irregular structures. A noteworthy observation is the marginally lower Kappa coefficient of MSSI-Net compared with CTIDRNet. This divergence arises from divergent architectural priorities: CTIDRNet employs deformable convolutions to optimize the spatial consistency in homogeneous regions, whereas MSSI-Net emphasizes multiscale discriminative capability for change-sensitive features, thereby achieving superior performance for practical CD applications.

TABLE I
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE LEVIR-CD DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

	LEVIR-CD					
Method	OA	IoU	F1	Pre	Rec	Kappa
ISNet [86]	98.80	78.70	88.34	90.24	86.50	87.90
DMINet [52]	99.07	82.99	90.71	92.52	88.96	87.57
LSNet [87]	99.01	82.34	90.77	**92.63**	88.98	86.57
SNUNet [13]	98.82	78.36	87.94	89.25	86.66	89.02
ChangeFormer [58]	99.04	82.66	90.50	90.83	**90.18**	89.53
SwinSUNet [27]	98.41	81.19	89.62	91.73	87.60	80.77
BIT [57]	98.98	81.72	89.94	90.33	89.56	89.45
ICIF [28]	98.95	80.97	89.48	90.87	88.14	88.50
BiFA [88]	99.06	82.96	90.69	91.25	89.86	89.26
DDRL [89]	**99.32**	**84.74**	**91.65**	**93.31**	90.17	**89.92**
CTIDRNet [90]	**99.22**	**83.59**	**91.06**	91.54	**90.59**	**90.78**
Ours	**99.39**	**85.21**	**92.15**	**93.68**	**90.67**	**90.33**

To confirm the validity of our proposed MSSI-Net on the LEVIR-CD dataset more intuitively, we select several samples and visualize their CD maps. Fig. 5(1) depicts a CD scenario featuring a solitary building against an intricate background. With the exception of ISNet and SNUNet, which exhibit varying degrees of missed detections resulting from insufficient sensitivity to subtle boundary variations, all other methods provide relatively comprehensive detection results. As shown in Fig. 5(3), SwinSUNet (SSUNet) demonstrates insuf-

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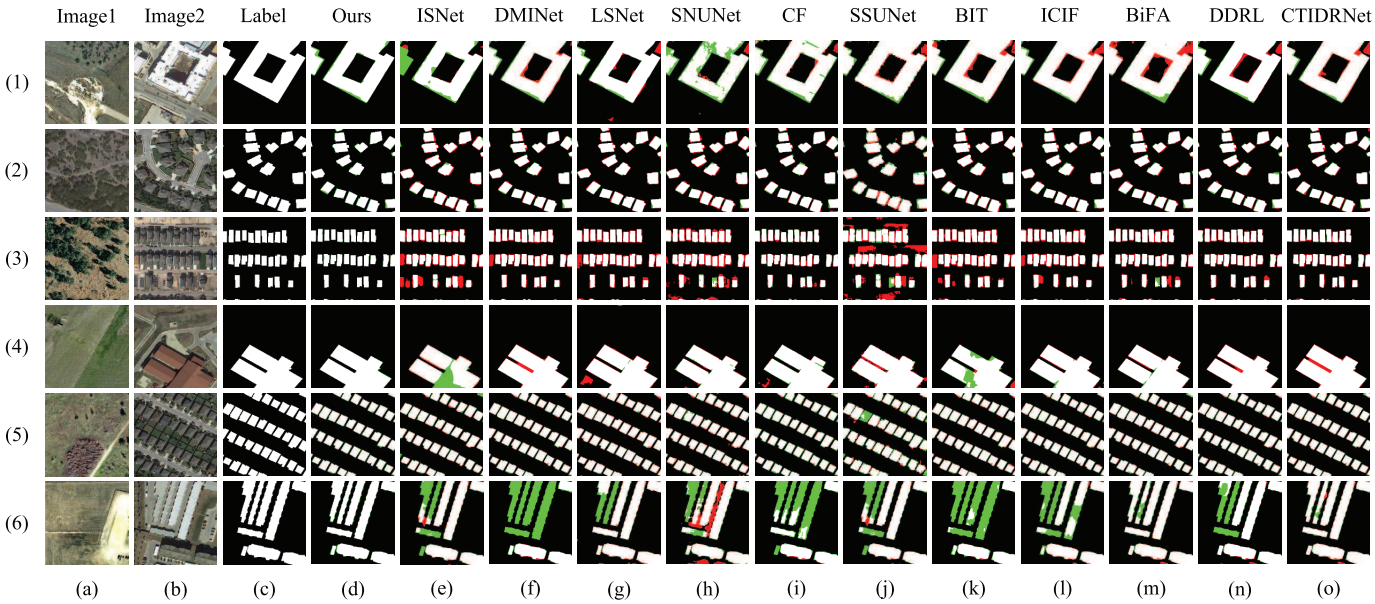


Fig. 5. Visualization results of different methods on the LEVIR-CD dataset. (1)–(6) Six pairs of RS images from the LEVIR-CD. (a) and (b) Input pretemporal and posttemporal images. (c) Ground truth. (d)–(o) Change maps of MSSI-Net, ISNet, DMINet, LSNet, SNUNet, ChangeFormer, SwinSUNet, BIT, ICIF-Net, BiFA, DDRL, and CTIDRNet. [The rendered colors represent true positives (white), true negatives (black), false positives (green), and false negatives (red).]

ficient accuracy in identifying large areas with notable false negatives and missed detections. Conversely, ChangeFormer and CTIDRNet reduce omissions via deformable convolutions but some false detections. Our MSSI-Net, in contrast, yields the most discernible results through manifold optimization of differential features, especially with clearer building edges compared with other methods. Fig. 5(4) shows a complex irregular building object, where BIT generates fragmented predictions, reflecting limitations in long-range dependency modeling. In comparison, ICIF-Net and BiFA are able to capture features more accurately; yet, they still fail to precisely reflect the actual situation of the changed areas. Our MSSI-Net delivers pixel-accurate delineation, empowered by high-order semantic guidance and multigranularity interaction. Fig. 5(2) and (5) reveals the CD in scenarios with densely packed buildings. Relatively complete CD results were obtained for all methods. However, our proposed method excels in reducing noise and pseudo-changes while maintaining the internal cohesion of the changed areas, producing results that are closest to the ground truth. This stems from the design of our CFFM module, which fosters the fusion of global semantic context and local texture embeddings, empowering the network to achieve a deep comprehension and perception of both intricate local details and the global information. These visual comparisons substantiate that our proposed MSSI-Net architecture establishes superior robustness against scenarios involving scale variations, occlusions, and pseudochanges.

2) Experimental Comparisons and Analysis on the CDD Dataset: The performance across methodologies on the CDD dataset is presented in Table II, this dataset records cropland utilization changes with a lower resolution and involves more complex scenarios. From this table, it is evident that BIT excels in global context modeling, scoring an $F1$-score of 97.36% and IoU of 96.80%, respectively. Furthermore, while DMINet and LSNet leverage intertemporal attention mechanisms, their limited multiscale fusion capability results in decreased recall metric. ChangeFormer achieves an $F1$ of 95.04% and IoU of 94.32%, highlighting the effectiveness of its advanced attention mechanism in enhancing feature information. In addition, DDRL outperforms most of the aforementioned networks with $F1$ and IoU reaching 97.62% and 95.55%, respectively, indicating that graph attention-enhanced domain learning can improve the accuracy of change information detection. The strength of MSSI-Net derives from two key components: the designed MIM module, which

TABLE II
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE CD DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

	CDD					
Method	OA	IoU	F1	Pre	Rec	Kappa
ISNet [86]	98.78	90.12	94.80	95.18	94.43	74.69
DMINet [52]	99.07	92.44	96.07	95.93	96.22	77.15
LSNet [87]	98.55	93.61	94.01	93.84	94.19	76.46
SNUNet [13]	98.50	89.57	94.49	94.83	94.16	70.99
ChangeFormer [58]	98.16	94.32	95.04	95.82	94.27	55.98
SwinSUNet [27]	98.50	88.63	93.97	95.70	92.30	65.12
BIT [57]	99.05	**96.80**	97.36	97.29	**97.44**	65.00
ICIF [28]	95.82	85.76	92.33	88.62	96.37	63.27
BiFA [88]	99.18	94.45	96.37	96.68	96.32	75.33
DDRL [89]	**99.59**	95.55	**97.62**	**97.98**	97.27	**79.84**
CTIDRNet [90]	**99.62**	**95.70**	**97.80**	**98.00**	**97.40**	**81.22**
Ours	**99.69**	**95.87**	**97.91**	**98.02**	**97.81**	**84.34**

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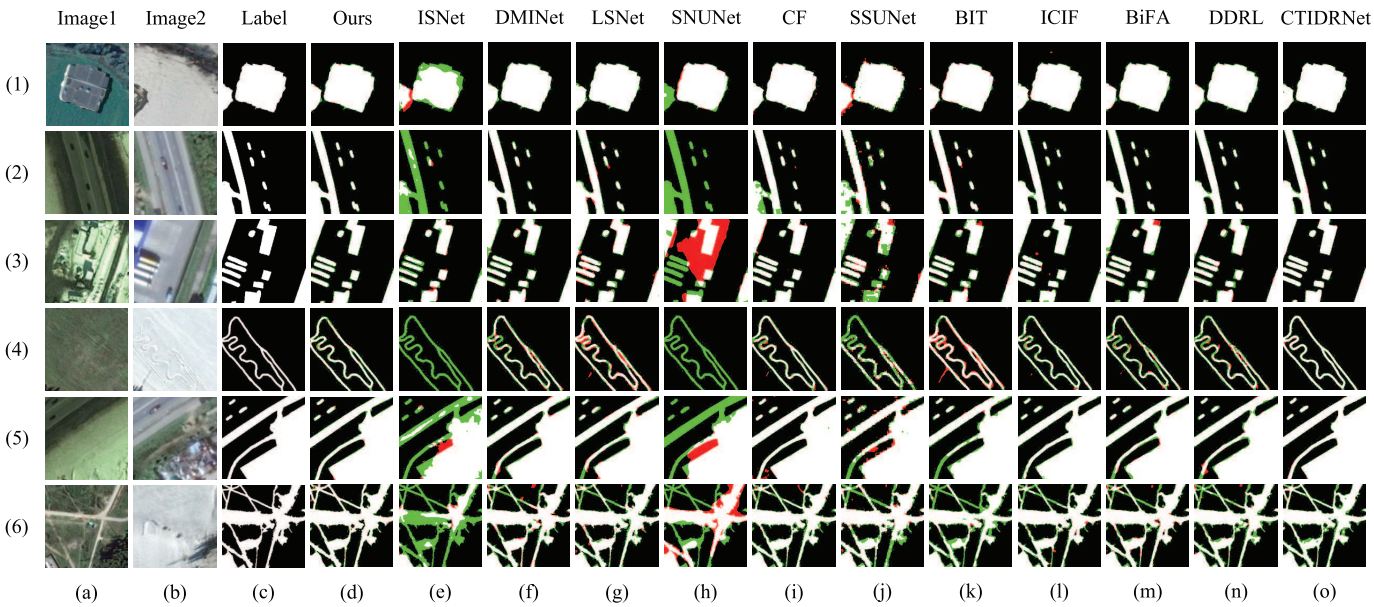


Fig. 6. Visualization results of different methods on the CDD dataset. (1)–(6) Six pairs of RS images from the CDD. (a) and (b) Input pretemporal and posttemporal images. (c) Ground truth. (d)–(o) Change maps of MSSI-Net, ISNet, DMINet, LSNet, SNUNet, ChangeFormer, SwinSUNet, BIT, ICIF-Net, BiFA, DDRL, and CTIDRNet. [The rendered colors represent true positives (white), true negatives (black), false positives (green), and false negatives (red).]

achieves precise localization of irregular crop boundaries through multiscale feature enhancement, while the CFFM module suppresses illumination-induced false changes through alignment of semantic cues with visual embeddings. Consequently, our proposed MSSI-Net attains peak performance in most metrics when confronted with complex scenarios, achieving OA at 99.69%, $F1$ at 97.91%, recall at 97.81%, precision at 98.02%, Kappa at 84.34%, and only 0.93% below BIT on the IoU metric. Notably, despite MSSI-Net slightly trailing BIT in the IoU metric, this discrepancy originates because BIT emphasizes pixel-level consistency through a pure transformer architecture, whereas MSSI-Net prioritizes preserving the integrity of changing regions to mitigate detrimental effects caused by spurious variation.

The visualization results of several representative samples from the CDD dataset are exhibited in Fig. 6. By comparing the outcomes of Fig. 6(1) with the rest of the content, it can be concluded that due to limited multiscale perception, ISNet can only deal with simple change objects and is deficient in the capacity to obtain complete regions in complex environments. While SSUNet and SNUNet show partial improvements in this area, they still struggle to excel in finding small-sized change regions in complex scenarios. Conversely, MSSI-Net in the case achieves seamless detection through semantic prompts, suppressing pseudo-changes induced by seasonal tillage. The CD results for winding roads, as illustrated in Fig. 6(4), manifest that BIT and LSNet are capable of detecting small object changes while CTIDRNet leverages deformable convolutions to improve small-object detection, enabling pixel-accurate localization of subtle changes. However, the loss of local details that occurs during the downsampling process poses challenges for them in maintaining precise boundaries and results in a tendency toward false detections. Moreover, as can be seen from the results of intricate surface change detection in Fig. 6(6), the ChangeFormer and BiFA truly reduce several false detections compared with the above results. Nevertheless, due to the lack of fine-grained details and local contextual guidance necessary for accurate localization, their detection results remain relatively coarse and incomplete when compared to the ground truth. Unlike other methods, our proposed method effectively integrates discrepancy and feature information to yield highly discriminative results. This is primarily attributed to the designed CFFM, which suppress the effect of variations such as season and illumination, in addition to our MDPH module, which strengthens the intraclass compactness of the differential space and reduces boundary ambiguities thereby ensuring accurate detection.

3) Experimental Comparisons and Analysis on the SYSU-CD Dataset: Table III summarizes the measures of the evaluation metrics obtained by all methods on the SYSU-CD dataset. It is noteworthy that all scores of the models in the SYSU-CD dataset are lower than those in the LEVIR-CD, CDD, and WHU-CD datasets. This is due to the fact that the SYSU-CD dataset contains complex variable objects such as roads, ships, and buildings, which are characterized by high diversity and complexity. Consequently, the detection results are more prone to being affected by noise, including factors like illumination, shadows, occlusion, and motion blur, leading to a higher number of false detections. Specifically, ICIF demonstrates suboptimal performance, primarily due to the inherent deficiency in its sequential feature aggregation mechanism, which fails to incorporate dynamic cross-scale interactions. This architectural constraint becomes particularly pronounced when confronting the complexities inherent to urban environments, where multiscale dependencies are critical for accurate detection. In addition, ChangeFormer employs pure self-attention for context modeling, yielding improved results. However, its difficulty in

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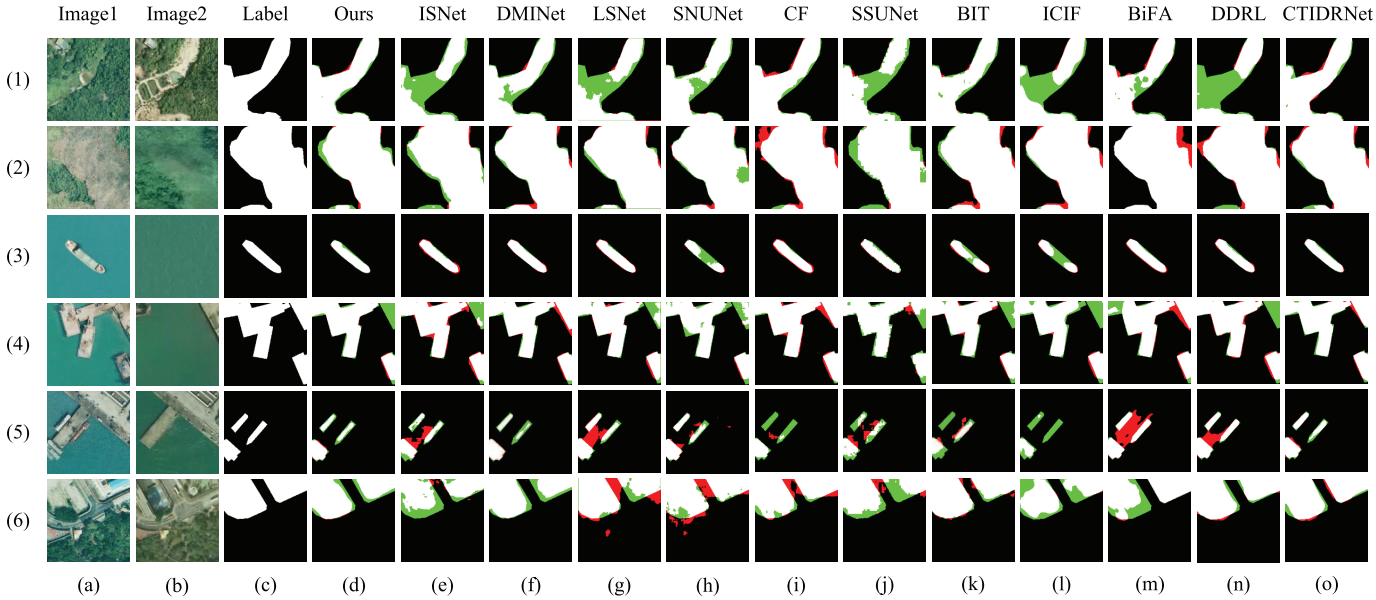


Fig. 7. Visualization results of different methods on the SYSU-CD dataset. (1)–(6) Six pairs of RS images from the SYSU-CD. (a) and (b) Input pretemporal and posttemporal images. (c) Ground truth. (d)–(o) Change maps of MSSI-Net, ISNet, DMINet, LSNet, SNUNet, ChangeFormer, SwinSUNet, BIT, ICIF-Net, BiFA, DDRL, and CTIDRNet. [The rendered colors represent true positives (white), true negatives (black), false positives (green), and false negatives (red).]

TABLE III

QUANTITATIVE COMPARISON RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE SYSU-CD DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

	SYSU-CD					
Method	OA	IoU	F1	Pre	Rec	Kappa
ISNet [86]	91.32	68.93	79.82	79.76	**79.87**	57.11
DMINet [52]	89.02	55.31	71.23	81.73	63.12	39.90
LSNet [87]	89.61	62.65	78.47	83.06	74.36	38.67
SNUNet [13]	89.30	61.75	77.76	82.45	73.56	39.03
ChangeFormer [58]	89.26	65.36	77.24	80.47	74.25	45.01
SwinSUNet [27]	87.46	61.24	75.95	80.53	71.87	35.59
BIT [57]	88.47	58.82	74.07	78.81	69.87	45.59
ICIF [28]	89.61	62.16	76.67	81.53	72.35	43.75
BiFA [88]	**94.37**	**70.38**	82.33	85.67	79.79	**60.74**
DDRL [89]	93.61	70.05	**82.38**	**87.09**	78.16	50.50
CTIDRNet [90]	**95.05**	**71.23**	**83.43**	**86.92**	**80.24**	**62.18**
Ours	**95.41**	**71.41**	**83.65**	**87.16**	**80.41**	**64.27**

harmonizing global semantics with local details thereby limits further performance enhancement. Furthermore, CTIDRNet achieves promising performance on this dataset through the unique attention mechanism. However, its excessive reliance on local feature refinement compromises global contextual modeling, leading to $F1$ and IoU scores that are 0.22% and 0.18% lower than our proposed MSSI-Net, respectively. Among these models, MSSI-Net showcases its superior performance in all metrics with OA, IoU, $F1$, precision, recall, and Kappa being 95.41%, 71.41%, 83.65%, 87.16%, 80.41%, and 64.27%, respectively. It can be seen from the results that our network also performs superiorly in handling complex and dynamic datasets. This is mainly attributed to our designed MIM module, which harnesses multiscale feature information to guide the network in perceiving multisource discriminative features. This ensures that the network possesses a robust detection capability when dealing with samples exhibiting complex variations.

The performance of RSCD models in predicting changes in bitemporal images from the SYSU-CD dataset is shown in Fig. 7. Compared with the other three datasets, the SYSU-CD dataset exhibits a diversity of scales in its targets, and the abundance of distinct categories among these targets elevates the level of complexity and challenge. Fig. 7(1) shows that LSNet and ISNet exhibit fragmented detections, struggling to reconcile slender road features due to their shallow multiscale fusion mechanisms. While the performance of ChangeFormer has been further improved, there are still considerable instances of undetected objects, as illustrated in the second-to-last row of Fig. 7(4). In addition, as shown in Fig. 7(2), CTIDRNet captures finer environmental details compared with SSUNet. However, its inherent emphasis on localized feature refinement inadvertently produces edge anomalies, particularly along geometrically complex boundaries. In most cases, DMINet can capture a wide area of changes, including variations in roads and environments. However, as depicted in Fig. 7(5), it struggles with identifying changes that exhibit subtle differences in appearance. Moreover, DDRL reduces false alarms through graph attention-enhanced domain learning, as seen in Fig. 7(6), where it suppresses illumination-induced pseudo-changes better than ChangeFormer. However, its oversmoothing of transitional zones between roads and buildings reveals limitations in handling intraclass variance, a chal-

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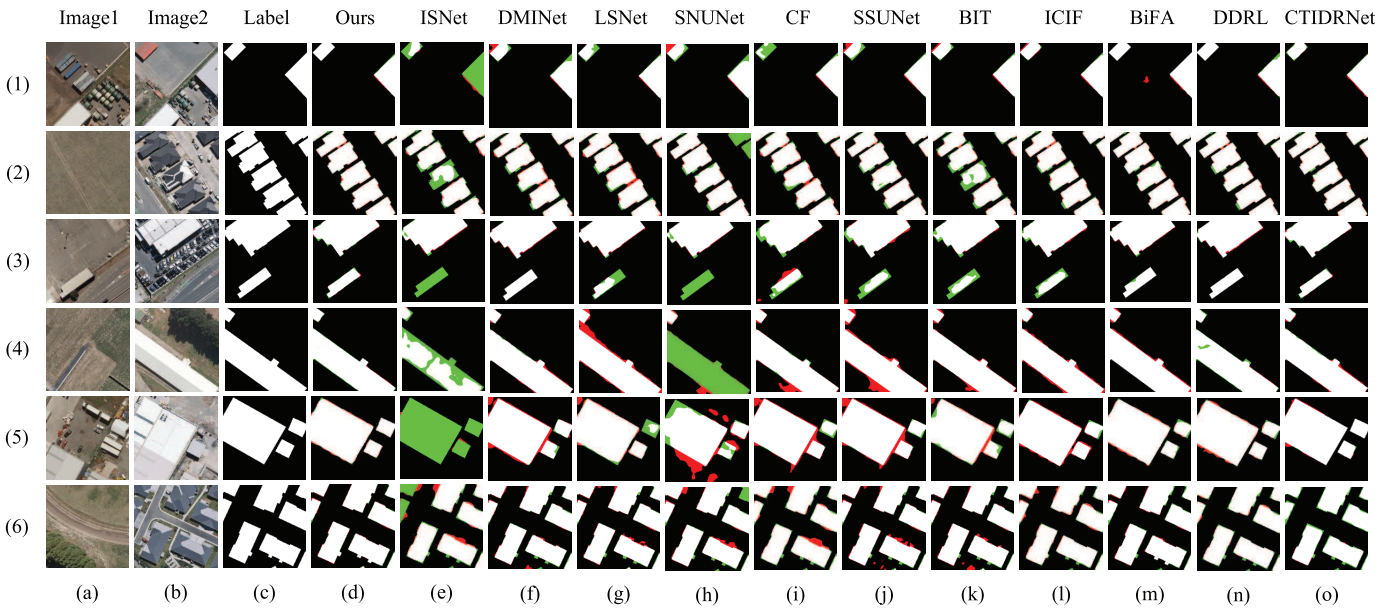


Fig. 8. Visualization results of different methods on the WHU-CD dataset. (1)–(6) Six pairs of RS images from the WHU-CD. (a) and (b) Input pretemporal and posttemporal images. (c) Ground truth. (d)–(o) Change maps of MSSI-Net, ISNet, DMINet, LSNet, SNUNet, ChangeFormer, SwinSUNet, BIT, ICIF-Net, BiFA, DDRL, and CTIDRNet. [The rendered colors represent true positives (white), true negatives (black), false positives (green), and false negatives (red).]

lenge exacerbated by diverse urban structures. Ultimately, it is evident that our proposed MSSI-Net stands out in detecting change features with reduced false positives and missed detections, while simultaneously preserving change boundary information. This superior performance is due to the discriminative feature extraction capability of MDPH and the noise interference suppression provided by the CFFM module.

4) Experimental Comparisons and Analysis on the WHU-CD Dataset: Table IV presents the quantitative performance evaluation results of the proposed MSSI-Net and other comparative methods on the WHU-CD dataset. This dataset primarily focuses on CD scenarios involving densely distributed, irregularly shaped, and long-range building clusters. As demonstrated, our network achieves outstanding performance across all key metrics, including OA, IoU, $F1$, precision, recall, and Kappa, with specific values reaching 99.58%, 90.15%, 94.82%, 95.63%, 94.02%, and 90.35%, respectively. While the sequential feature concatenation of ISNet degrades the recall by 7.48% compared with MSSI-Net. Also, LSNet achieves balanced precision–recall tradeoffs but lacks robustness under occlusion-heavy scenarios. Furthermore, DDRL employs graph attention to enhance domain separability but struggles with intraclass variance in densely packed buildings, as evidenced by its lower recall of 90.58%. In addition, BiFA achieves competitive metrics results such as 94.37% for $F1$ and 89.34% for IoU through bitemporal channel–spatial alignment, but it exhibits limitations in global context modeling because its alignment module prioritizes local interactions over holistic scene understanding. Compared with other methods, our MSSI-Net gains better results through the MIM module, providing highly discriminative multigranular features for subsequent cross-modal alignment.

TABLE IV
QUANTITATIVE COMPARISON RESULTS OF DIFFERENT CHANGE DETECTION METHODS ON THE WHU-CD DATASET. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

	WHU-CD					
Method	OA	IoU	F1	Pre	Rec	Kappa
ISNet [86]	94.90	64.37	78.62	72.03	86.54	81.74
DMINet [52]	99.22	81.87	89.93	92.44	87.56	89.68
LSNet [87]	98.56	83.55	90.74	88.05	**93.60**	87.91
SNUNet [13]	96.41	70.61	82.77	75.03	92.31	85.63
ChangeFormer [58]	99.26	82.32	90.30	93.73	87.11	88.18
SwinSUNet [27]	96.90	70.37	82.61	80.31	85.04	**90.12**
BIT [57]	98.08	75.00	85.71	82.04	89.74	89.34
ICIF [28]	99.01	80.43	89.16	90.79	87.58	89.25
BiFA [88]	**99.56**	**89.34**	**94.37**	**95.15**	**93.60**	**90.90**
DDRL [89]	99.08	84.20	91.43	92.30	90.58	88.05
CTIDRNet [90]	**99.32**	**86.07**	**92.52**	**94.18**	90.91	88.32
Ours	**99.58**	**90.15**	**94.82**	**95.63**	**94.02**	**90.35**

The MDPH module, on the other hand, further improves the performance of our network on this dataset by integrating multilevel semantic guidance to harmonize spectral, geometric, and temporal discriminative features.

The visualization results in Fig. 8 highlight the CD performance of the various comparison methods as well as our proposed MSSI-Net in challenging urban scenarios on the WHU-CD dataset. Specifically, in a dense cluster of residential buildings as shown in Fig. 8(2), ICIF generates

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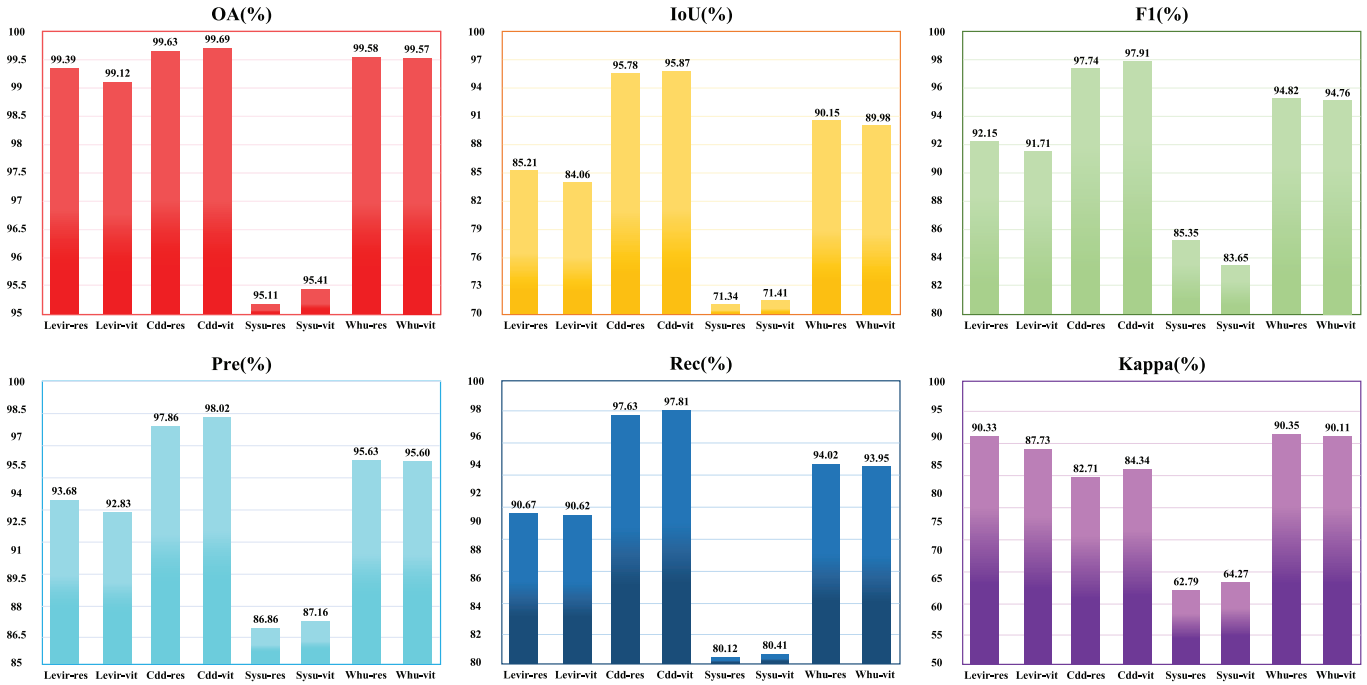


Fig. 9. Ablation experiments based on different backbones and scores for various metrics.

coherent detections but introduces false positives between adjacent structures due to incomplete global attention, while ISNet oversmooths transitional zones, misclassifying connected rooftops as unchanged. In addition, SwinSUNet is unable to completely detect irregularly shaped buildings as shown in Fig. 8(3), due to its rigid window segmentation, while our proposed MSSI-Net precisely delineates complex building contours, suppressing noise from reflective surfaces. Furthermore, CTIDRNet captures fine architectural details in elongated regular buildings presented in the species of Fig. 8(4) through deformable convolutions but produces jagged boundaries, reflecting its local refinement bias, whereas MSSI-Net maintains smooth boundaries of elongated features without omissions and misdetections.

As demonstrated in Fig. 8(5) and (6), we can conclude that BiFA achieves commendable detection without large-scale leakage and misdetection. However, a discernible limitation lies in its insufficient precision regarding the delineation of building edges, manifesting as coarse false-negative artifacts. This deficiency stems from its design of philosophy prioritizing global alignment robustness over local accuracy. Consequently, in geometrically intricate urban scenarios, this approach becomes susceptible to edge artifacts and missed detections, particularly where architectural boundaries exhibit high variability and complexity. Comparatively, our designed MSSI-Net consistently delivers pixel-accurate results across diverse scenarios. This enhanced performance originates from the MIM module, which dynamically fuses shallow details and deep semantic contexts, solves the leakage detection problem of narrow gaps, and refines the delineation of irregular building boundaries. Moreover, the MDPH module further strengthens the intraclass compactness of the differential feature space and effectively minimizes fragment detection in complex urban layouts.

TABLE V
EFFICIENCY COMPARISON RESULTS OF DIFFERENT METHODS

Method	Params (M)	FLOPs (G)
ISNet [86]	14.36	9.58
DMINet [52]	6.76	14.55
LSNet [87]	1.10	8.53
SNUNet [13]	12.03	27.44
ChangeFormer [58]	41.03	74.46
SwinSUNet [27]	27.24	10.72
BIT [57]	3.01	8.52
ICIF [28]	25.83	25.41
BiFA [88]	9.87	53.00
DDRL [89]	31.18	42.08
CTIDRNet [90]	78.64	55.70
Ours	**30.49**	**41.11**

F. Ablation Study

In order to ascertain the effectiveness of our proposed MSSI-Net, the requisite ablation experiments will be conducted in this section. Table V verify the efficiency of the proposed model, Table VI and Fig. 9 illustrate the impact of utilizing distinct backbones, while Table VII validates the utility and reliability of the diverse modules devised in this study. Figs. 10–13, substantiate the exceptional capacity of the network to address change characteristics as manifested in thermal feature maps. Fig. 14 further validates the impact of semantic guidance on CD performance.

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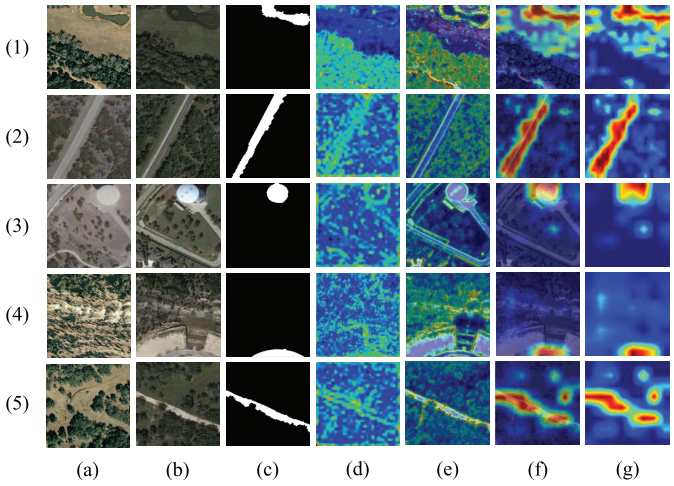


Fig. 10. Visualization results for semantic role analysis on the LEVIR-CD dataset. (1)–(5) Five pairs of images of LEVIR-CD. (a) and (b) Input bitemporal images. (c) Ground truth. (d)–(g) Change area feature maps at the stage of Baseline, +MIM, +CFFM, and +MDPH.

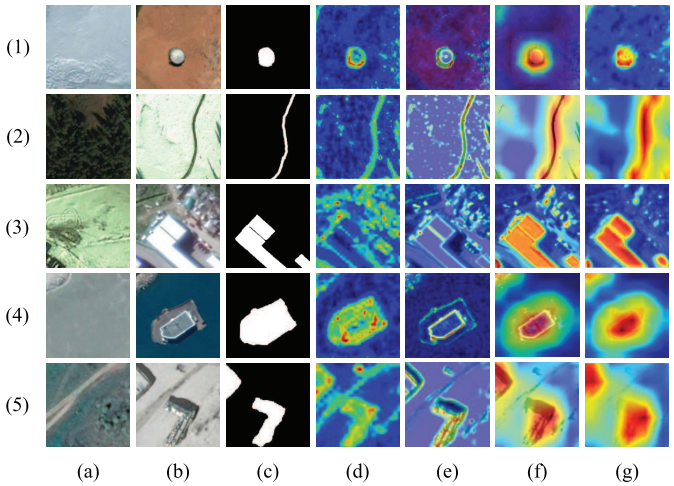


Fig. 11. Visualization results for semantic role analysis on the CDD dataset. (1)–(5) Five pairs of images of CDD. (a) and (b) Input bitemporal images. (c) Ground truth. (d)–(g) Change area feature maps at the stage of Baseline, +MIM, +CFFM, and +MDPH.

1) Analysis of Parameters and Running Efficiency: To assess the practical efficiency of the proposed method, we compare its parameter size (Params) and floating-point operations per second (FLOPs) with 11 state-of-the-art CD methods, where the input size is $256 \times 256 \times 3$. As shown in Table V, our method demonstrates lower FLOPs (41.11G) than recent approaches like BiFA and DDRL, validating the efficiency of our architectural design. While our Params (30.49M) are significantly smaller than transformer-based models such as ChangeFormer and CTIDRNet, it remains larger than lightweight architectures such as BIT and LSNet. This reflects the computational demands introduced by our MIM and CFFM modules, which are designed to prioritize the feature discriminability. However, the inherent challenge of achieving semantic-visual alignment across multiscale representations and heterogeneous modalities necessitates significant computational resources, indicating potential avenues for future optimization.

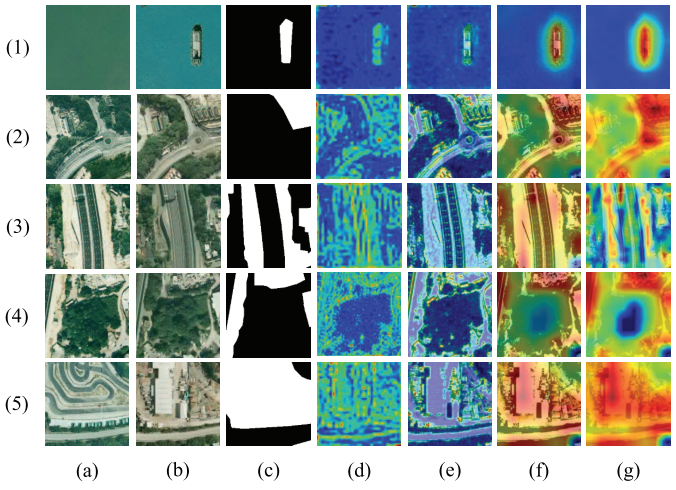


Fig. 12. Visualization results for semantic role analysis on the SYSU-CD dataset. (1)–(5) Five pairs of images of SYSU-CD. (a) and (b) Input bitemporal images. (c) Ground truth. (d)–(g) Change area feature maps at the stage of Baseline, +MIM, +CFFM, and +MDPH.

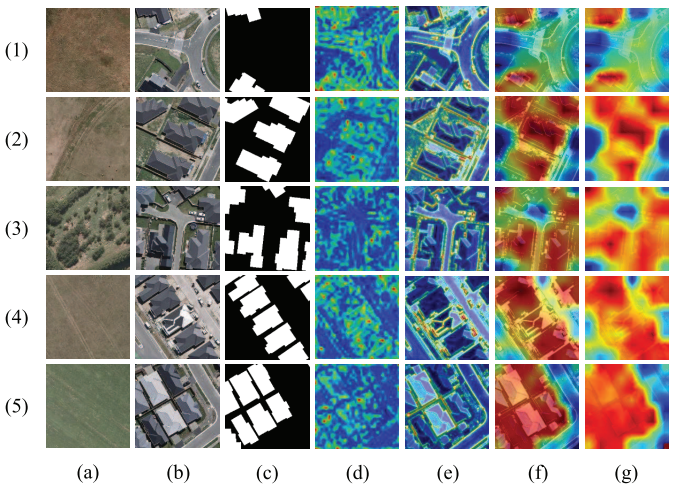


Fig. 13. Visualization results for semantic role analysis on the WHU-CD dataset. (1)–(5) Five pairs of images of WHU-CD. (a) and (b) Input bitemporal images. (c) Ground truth. (d)–(g) Change area feature maps at the stage of Baseline, +MIM, +CFFM, and +MDPH.

2) Comparative Ablation Across Different Backbones: To evaluate the effectiveness of our proposed network, we conduct ablation experiments on the LEVIR-CD, CDD, SYSU-CD, and WHU-CD datasets for two skeleton networks, namely RN-50 (ResNet-50) and ViT (ViT-B/16), which are embedded within the CLIP model. As illustrated in Table VI, the performance of ViT and RN-50 exhibits variability across different datasets. Notably, ViT demonstrates superior performance compared to RN-50 on both the CDD and SYSU-CD datasets. Specifically, in terms of the OA metric, ViT outperforms RN-50 by 0.06% and 0.30%, respectively. This can be attributed to the fact that ViT has a more robust global receptive field, which allows it to capture global correlations among individual objects in RS images more effectively. Furthermore, the image sequence features produced by the ViT backbone align more closely with text sequence features, resulting in stronger performance in multimodal tasks. This is evident in the $F1$ metric, where ViT surpasses RN-50 by 0.17% and 0.30% on the two datasets, respectively.

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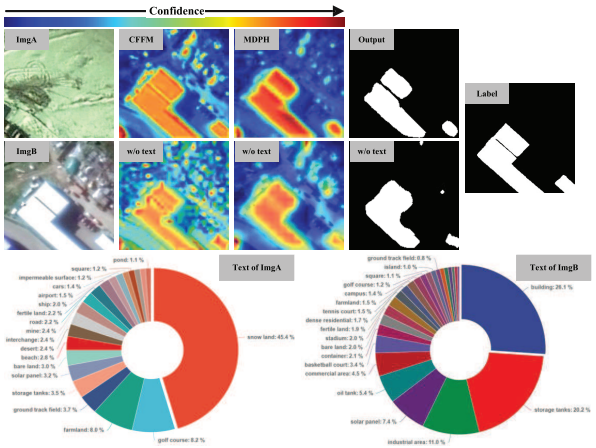


Fig. 14. Visualization comparison of change masks with/without textual guidance.

TABLE VI

ABLATION EXPERIMENTS ON DIFFERENT BACKBONES. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

Method	Backbone	OA	IoU	F1	Pre	Rec	Kappa
LEVIR-CD	**Resnet-50**	**99.39**	**85.21**	**92.15**	**93.68**	**90.67**	**90.33**
	ViT-B/16	99.12	84.06	91.71	92.83	90.62	87.73
CDD	Resnet-50	99.63	95.78	97.74	97.86	97.63	82.71
	ViT-B/16	**99.69**	**95.87**	**97.91**	**98.02**	**97.81**	**84.34**
SYSU-CD	Resnet-50	95.11	71.34	83.35	86.86	80.12	62.79
	ViT-B/16	**95.41**	**71.41**	**83.65**	**87.16**	**80.41**	**64.27**
WHU-CD	**Resnet-50**	**99.58**	**90.15**	**94.82**	**95.63**	**94.02**	**90.35**
	ViT-B/16	99.57	89.98	94.76	95.60	93.95	90.11

However, on the LEVIR-CD and WHU-CD datasets, it is observed that the performance of ViT-B/16 is slightly inferior to RN-50. The gaps in the IoU metric reach 1.15% and 0.17%, respectively. This is because the LEVIR-CD and WHU-CD datasets comprise more variable local targets, posing challenges for the ViT model. Conversely, ResNet-50 adopts a technique of progressively extracting image features via convolutional layers. Its local receptive fields and hierarchical architecture allow for a more efficient capture of intricate details and multiscale information within images, ultimately yielding superior performance outcomes. In summary, based on the comparison presented inthe bar chart of Fig. 9, we can conclude that the two types of backbones exhibit comparable capability on both the LEVIR-CD, CDD, and WHU-CD datasets. Despite a slight decline in our performance on the SYSU-CD, attributable to the inherent complexity and diversity of the dataset. The two types of backbones employed in our proposed MSSI-Net have achieved impressive results when compared with other methods, validating the advanced nature and robustness of our method in CD tasks.

3) Comparative Ablation Across Different Modules: To validate the effectiveness of the modules in the proposed MSSI-Net, we implement ablation studies on the LEVIR-CD, CDD, SYSU-CD, and WHU-CD datasets with varying settings based on the premise of cross-modal fusion. As demonstrated

TABLE VII

ABLATION EXPERIMENTS ON DIFFERENT MODULES. ALL RESULTS ARE DESCRIBED IN PERCENTAGE (%)

Method	Modules	OA	IoU	F1	Pre	Rec	Kappa
LEVIR-CD	Baseline	98.03	83.78	90.63	91.55	89.74	76.70
	w/o MIM	98.50	85.03	91.68	93.16	90.24	78.07
	w/o MDPH	98.85	84.22	91.18	92.04	90.33	85.55
	MSSI-Net	**99.39**	**85.21**	**92.15**	**93.68**	**90.67**	**90.33**
CDD	Baseline	98.97	93.35	96.66	96.99	96.34	65.87
	w/o MIM	99.47	94.21	97.54	97.20	**97.88**	81.03
	w/o MDPH	99.43	95.67	97.70	97.86	97.55	73.36
	MSSI-Net	**99.69**	**95.87**	**97.91**	**98.02**	97.81	**84.34**
SYSU-CD	Baseline	90.27	70.01	81.90	81.91	79.25	46.23
	w/o MIM	93.84	70.89	82.96	86.64	79.58	53.89
	w/o MDPH	94.53	71.22	83.33	**87.29**	79.71	57.00
	MSSI-Net	**95.41**	**71.41**	**83.65**	87.16	**80.41**	**64.27**
WHU-CD	Baseline	98.85	89.10	94.00	93.95	93.00	81.06
	w/o MIM	99.30	89.85	94.65	95.25	93.85	84.51
	w/o MDPH	99.42	89.98	94.70	**95.70**	93.95	86.42
	MSSI-Net	**99.58**	**90.15**	**94.82**	95.63	**94.02**	**90.35**

in Table VII, the proposed MSSI-Net achieves the optimal results across all metrics on the LEVIR-CD dataset. It outperforms the baseline by 1.36%, 1.43%, 1.52%, 2.13%, 0.93%, and 13.63% in the metrics of OA, IoU, $F1$, precision, recall, and Kappa, respectively.

In addition, the results indicate that the incorporation of the MIM or MDPH, either individually or in conjunction with other modules, can also increase various indicators relative to the baseline, thereby substantiating the efficacy of the designed MIM and MDPH in improving the performance of CD on this specific dataset. In addition, upon examination of the CDD dataset, we find that the addition of MIM to the baseline method results in notable improvements in performance. Specifically, the $F1$-score and IoU value increase by 0.88% and 0.86%, respectively. These enhancements exhibit the effectiveness of MIM in promoting the performance of the baseline method. Furthermore, the incorporation of the MDPH results in a considerable elevation in the performance of our network, with the $F1$ and IoU metrics reaching 97.91% and 95.87%, respectively. It is relevant to acknowledge that although the elimination of the MIM module may provide a modest advancement in recall as well, its impact on other metrics demonstrates the indispensable role in optimizing detection performance. Furthermore, the SYSU-CD dataset reveals that the MSSI-Net achieves optimal performance across all metrics, including OA, $F1$, IoU, recall, and Kappa. However, upon removing the MDPH component, an enhancement in Precision can be observed, with an increase of 0.13% compared with its previous value. This is due to the fact that the MDPH offers a comprehensive perception of variational information. Nevertheless, when the model focuses on unnecessary or redundant details, it adversely impacts the performance to accurately discern genuinely relevant information. However, the overall results show that the full MSSI-Net remains the

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superior choice, which confirms the efficiency of the module in augmenting the overall performance.

Moreover, the ablation experiments conducted on the WHU-CD dataset demonstrate that the complete MSSI-Net achieves optimal performance across all evaluation metrics, outperforming the baseline model by 0.73%, 1.05%, 0.82%, 1.68%, 1.02%, and 9.2% in the OA, IoU, $F1$, precision, recall, and Kappa metrics, respectively. Conversely, it can be observed that ablating the MIM module decreases IoU by 0.30% and $F1$ by 0.17%, indicating that it plays the essential role in refining the multiscale feature interactions, thereby improving detection completeness. In contrast, the absence of the MDPH module produces a marginal increase in Pre but incurs a substantial decline in the Kappa coefficient, suggesting that MDPH not only strengthens predictive consistency but also effectively balances the precision–recall tradeoff. Collectively, these results confirm that the modular design of MSSI-Net addresses the critical challenges in detection accuracy and robustness, particularly when dealing with complex urban building change scenarios. Ultimately, the ablation results presented in this section demonstrate the distinctive contribution of the modules proposed in this article. The synergistic interaction between MIM and MDPH ensures comprehensive feature refinement and adaptive scale perception, establishing the network's superiority in CD tasks. Furthermore, the higher precision exhibited by our proposed MSSI-Net, compared with the baseline, further verifies the superiority of our method.

4) Analysis of Semantic Roles via Visualization Ablation: To conduct a visual analysis of the influence of high-order semantics within the proposed methodology, we have generated heatmaps depicting the features at each stage. The results of these visualization experiments are displayed in Figs. 10–13. Specifically, "+MIM," "+CFFM," and "+MDPH" denote that after progressively adding each module, the output results of the feature maps at each stage. First, by comparing the visualization results obtained through the MIM module with those of the original baseline approach, we can conclude that the incorporation of the MIM component markedly boosts the OA of the feature images. Especially on the LEVIR-CD dataset shown in Fig. 10(3) and (4), it is apparent that the MIM module substantially refines the discriminability of the regions of interest, effectively mitigating background noise interference while preserving abundant detailed information. Furthermore, for the WHU-CD dataset, the MIM module enhances the detection of densely packed residential buildings by resolving ambiguities in overlapping rooftops and narrow gaps. For example, the boundaries of buildings and roads, the distinction between minute targets, and the variations stemming from environmental perturbations are shown in Fig. 13(4) and (5). This is attributed to the multiscale visual interaction process within the MIM module, where the generated high-order visual semantic representations optimize the ability to process visual information across hierarchical levels.

Moreover, the images depicted in the Fig. 12(2) and (5) intuitively illustrate the efficacy of our CFFM component on the SYSU-CD dataset. It is observable that targets such as building clusters and bridges, which originally lack clear indications of change areas, have received explicit attention after processing through the CFFM module. This enhancement stems from the inherent capability of CFFM to integrate multimodal visual-language features. The semantic information derived from cross-modal fusion serves as a potent guide for the network to discern and locate areas of change, thereby bolstering the overall effectiveness of CD. Ultimately, examining the results represented in the Fig. 11(3) and (5) on the CDD dataset, we can find that the previously scattered and diverted attention points have undergone successful aggregation and correction through MDPH. It mitigates fragmented predictions by enforcing manifold consistency in differential space, ensuring coherent detection of structural modifications while suppressing pseudo-changes from temporary obstructions. This confirms that the diversified fusion of differential features guided by vision-language information significantly improves the capacity to predict changes in bitemporal RS images. Also, the final predicted feature maps exhibit remarkable discriminability and precision, demonstrating the superiority of MSSI-Net in executing CD tasks.

5) Analysis of Textual Semantics via Visualization Ablation: Building upon the preceding experimental results, we conduct a further validation of the effectiveness and accuracy of text generated by RemoteCLIP. This validation explicitly illustrates the significance of the text modality within our proposed method and elucidates its role in enhancing the model's capability to understand change regions. As demonstrated in Fig. 14, a comparison between results obtained with and without the integration of high-confidence textual cues reveals distinct advantages for the text-guided branch. Specifically, its heatmaps and final predicted masks exhibit sharper building boundaries and more complete change representations. Conversely, the branch operating without textual modality assistance is demonstrably more susceptible to high-resolution noise interference and yields comparatively coarser predictions. This experiment further underscores the efficacy and superiority of our proposed MSSI-Net for RSCD through cross-modal fusion.

V. Conclusion

In this article, we introduce a multimodal network named MSSI-Net tailored for the RSCD task. To optimize the accuracy of the differential feature extraction process, we devise the MIM module to facilitate the perception of multiscale visual features. In addition, our proposed CFFM leverages higher order interactions to harness both homogeneous and heterogeneous features across modalities, thereby ensuring the generated vision-language features are semantically representative. Furthermore, we optimize the fusion of disparity features by incorporating the MDPH for the exploitation of multimodal information. Consequently, MSSI-Net is capable of generating precise and holistic representations, enhancing the overall performance in the RSCD task. To assess the effectiveness of MSSI-Net, we undertake extensive experiments across three benchmark datasets for CD: LEVIR-CD, CDD, and SYSU-CD. The experimental outcomes unequivocally confirm the superiority and efficacy of our proposed MSSI-Net in performing the RSCD task.

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